Weird Tales

The Unique Magazine

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EXPLORERS into INFINITY

By Ray Cummings.

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FOREWORD

Some of my present readers will doubtless remember "The Girl in the Golden Atom." When I wrote that book of the realm of infinite smallness there was in my mind its logical converse, the realm of the infinitely large. The one a complement to the other. And so I offer "Explorers Into Infinity," in no sense as a sequel to "The Girl in the Golden Atom," for fictionally they have no connection, but rather as its companion story.

You will find here a complete theory of the material universe as I conceive it may perhaps really be. To my own imagination—and I think very likely to your own—it is difficult to conceive of an infinite distance beyond the stars—empty Space stretching out forever. Nor is Einstein more satisfying to me, rather less so, for out beyond the Einstein system of curved Space must lie something or nothing. It is the nothingness which puzzles me. I have tried vainly to imagine a realm, infinitely large, of unending nothingness. Time is equally puzzling. I can conceive of eventful eons lying ahead of us; but rob that time of its future events and I flounder. To me at least, the conception of Time with nothing ever happening anywhere is impossible. To me also, an event presupposes the existence of something; and so, in my effort to imagine the infinitely large—Space illimitable, Time unending—I am forced to conceive what must fill that Space, what must happen to create that time.

You may call this tale fantastic, weird, bizarre. Doubtless it is. But with our most powerful microscopes reaching inward so tiny a distance to see no end in infinite smallness; our greatest telescopes groping futilely out into largeness unending to our vision, what is left but our imagination? And that, at least, we can send winging into the infinite!

I would not have you fear from this foreword that my story may be some pedantic, heavily technical exposition. It is not; for it is fiction only—a romance with which to entertain you; an effort, by using fictional methods, to reduce theories purely imaginative into concrete form with as great a degree of plausibility as may be. It is this only I desire: to carry you with me as you read; to make plausible this flight of our imaginations momentarily *set free from the tiny everyday universe which is all we have physically to envisage.*

RAY CUMMINGS.

CHAPTER 1

FREEDOM IN TIME AND SPACE

I was busy with the Martian mail which had just arrived when the message from Brett Gryce reached me. I did not apprehend that there was anything of secrecy about it, since he was using the open air; yet there was in his voice a note of tenseness and his summons was urgent.

"I can't come, Brett, until I get through the mail." I was rushed, and in a mood of ill-temper at the universe in general.

"When will that be?" he demanded.

"I don't know. It's accursedly large. Most of it seems to call for radio distribution—these Martians are always in a hurry."

"Come when you can," he said quietly.

"Tonight?"

"Yes—tonight. No matter how late—I must see you, Frank."

"I'll come," I said, and cut him off.

It was long past trinight, with dawn beginning to brighten the sky beyond the masonry of lower Great-New York, when I had disposed of those miserable Martian dispatches. The Gryces lived in the Southern Pennsylvania area. My aerocar was at hand. I had rather planned to use it; but I was tired and in no mood for effort. I decided to take the pneumatic, since there was a branch—little traveled, it is true—which would drop me within some twenty kilometers of the Gryce home.

They gave me an individual cylinder, with a bed if I cared to sleep. I did not. I lay there wondering what Brett could want of me; pleased also that I would see Francine—dear little Frannie. . . .

Occasionally I would call the Director ahead. They are sometimes careless in the switching of special individual cylinders; and I had no wish to pass the branch and find myself bringing up at some gulf terminal with half the morning getting back. Once I called Brett. He would meet me with his aero at the end of the branch when I arrived. He, too, reminded the Director. A surly sort of fellow; the Gryces had already reported him to the General Traffic Staff of Great-London.

I was not misdirected, however; but it was broad daylight when I emerged to find Brett impatiently awaiting me. And in a few minutes more we were landing at the aero-stage beside the Gryce home.

It was a simple enough place—for all Dr. Gryce's reputed wealth. An estate of a few kilometers, set in a heavy grove of trees with a high metallic wall about it. The granite house itself was small, unpretentious. There were few outbuildings; one a large rectangular affair which vaguely I understood was a workshop. I had never been in it. I knew old Dr. Gryce was interested in science; in his day he had materially advanced civilization with several fundamental devices. But what—if anything—he might be doing now, I had no idea.

Brett would tell me nothing beyond the fact that his father had suggested they send for me. But he seemed excited, tense. Dr. Gryce greeted me with his familiar kindliness. Though I did not see as much of this family as I would like (my business with the Interplanetary Mails was wholly underpaid and miserably confining), yet I counted the Gryces among my closest friends.

Dr. Gryce said, "We are very glad to see you, Frank. Come outside. Frannie is preparing breakfast."

His manner was grave and quiet as always. But there was about him also an air of tenseness; and an aspect of apprehension. And it struck me, a sort of weary, resigned depression which suddenly made his years sit more heavily upon him. He was a man of some eighty odd; and though for him no more than twenty or thirty years of life could be anticipated, I had never considered him really old. He was small, slight of frame, but erect, sturdy and vigorous. A smooth-shaven face with no more lines upon it than a keen intellect and a character once wholly forceful would engrave. And a mass of snow-white shaggy hair to make his head appear preternaturally large.

He seemed old now, however, with that sense of depression hanging upon him. And an indefinable aspect of fear.

I must allot a word to picture the three children of Dr. Gryce, motherless since childhood. Brett was now twenty-eight—three years older than myself, and physically my opposite. I am short, slender and rather dark. And—so they tell me—not too even of temper. Brett was a blond young giant. Crisp, wavy blond hair, blue eyes and the strong-featured, ruddy face of a handsome athlete. But not too handsome, for there was upon him no consciousness of his essentially masculine beauty. He was wonderfully good-natured. His was a ready, hearty laugh. He looked at life often from the humorous viewpoint. But he had also a touch of his father's grave dignity; and a keen intellect and a soberness of thought and reason far beyond his years.

The two other children—Martynn and Francine—were twins, now just seventeen. Alike, physically and temperamentally, as children of a birth traditionally should be. Slim and rather small—Martynn about my height; Francine somewhat shorter. Both blue-eyed, with blond hair. Francine's hair was long-waving tresses which she wore generally in plaits over her shoulders; Martynn's was short and curly. They were rather alike of feature; a delicacy of mold which gave to Martynn a girlishness. But not an effeminacy, for he was a young daredevil; and his sister hardly a lesser one. In childhood and adolescence an impish spirit of deviltry had always seemed to possess these twins; a spirit of mischief which had made them a great trial to their father. It had turned, now that they were nearing maturity, into an apparent desire for reckless adventure—the product of abounding health, and bubbling, irrepressible good nature. They adored each other; were constantly together, with youthful escapades threatening limb and life and complete disaster, out of which they would emerge or be extricated with dauntless spirits unperturbed.

The greater maturity of womanhood at seventeen had brought to Frannie moments of gentleness, sweetness and a simple dignity. But they were brief moments, and no more than a word or look from her twin was needed to dispel them. Martt himself was without a vestige of dignity. But they were no fools, these twins. They could, upon strict necessity, give sober, intelligent thought to any problem at hand (Martynn had won honors at the Great-London University); but of sober, matured action they were incapable. Fearless—unreasonably fearless. But irresistible, likable, and apparently quite capable of being restrained. A word from Dr. Gryce, or from Brett—and to a lesser extent from me who had known them from childhood—brought instant though often very temporary obedience. They considered themselves quite grown up now. In truth, at seventeen, Frannie was to my eyes a really beautiful young woman.

II

We sat in a little arbor beside the house, with its breakfast table already laid. Dr. Gryce, Brett, and myself. Martt was with Frannie preparing the meal. It was evidence of the simplicity which marked the Gryce household. In these days of mechanical devices for almost everything—and the usual multiplicity of servants—there was not a meal prepared for Dr. Gryce save by his daughter.

I was very curious to learn why they had sent for me; but I had no need to question, for at once Dr. Gryce plunged into it.

"I hope, Frank, that you can stay—well, at least a few days with us. Can you?"

I stared. The Day Officer of the Manhattan Interplanetary Postal Division was undoubtedly already in a rage at my absence. I said so. "A few days? Dr. Gryce, I dread every conjunction that brings these accursed mails—my divisional officers think it's a crime even to eat or sleep when a planet is near us."

He smiled. "I imagine I can fix it."

"Then I'll stay, of course. If you could fix the planetary orbits so that they were parabolas, Dr. Gryce, it would suit me exactly."

He and Brett both were smiling, but Dr. Gryce's smile was momentary, for at once that indefinable air of trouble returned to him.

"Frank," he said, "I hardly know how to begin telling you what we have done—are about to do. It seems curious also—I know it will strike you so, you have been such a friend to me and my children—that during all these years we have given you no hint of our purpose."

"We have told no one," Brett put in: "no one in the world."

I said nothing, but my curiosity increased. It was doubtless of grave import, this thing they had to tell me; the solemnity, earnestness which stamped them both was unmistakable.

For a moment Dr. Gryce was silent; then he said abruptly, "You know, Frank, all my life I have been engaged with science. In a measure, I have been successful; there are a few devices which will bear my name when I am gone."

I nodded. "I know that very well, Dr. Gryce."

"But all those things," he added earnestly, "all that I stand for to the world, has really been of little importance to me. My main labor, goal, dream, if you will, I have never told anyone—not a living person except my children. For ten years past Brett has been helping me. And though you would hardly believe it, for the last year or two Martt and Frannie have been of material aid in the accomplishment of my purpose."

"What branch of science?" I asked. "And you've accomplished it? You're ready to give it to the world?"

"Accomplished it—yes. But we are not ready to give it to the world—perhaps we never shall. There would be evil in it—evil diabolical—in untrained or unscrupulous hands. But we are ready to test it—a practical test. Tonight, Frank, my boy Brett is going upon an adventure——"

The fear which had been lurking in his eyes leaped to stamp his other features. He was afraid for Brett—afraid of this thing they were going to do. He had stopped abruptly; and more quietly he added:

"I want you to understand me, Frank, and so for a moment we must be wholly theoretical. This thing we are about to do involves the construction of our whole material universe. You know, of course, that no limit has been found to the divisibility of matter?"

His sudden question confused me. "You mean," I stammered, "that things can be infinitely small?"

"That there is no limit to smallness," Brett put in. "An atom—an electron—they are mere words. Within them conceivably might be a space with stars, planets, suns—worlds of their own so tiny that compared to the Space in which they roam that Space would seem—and would be—illimitable. Picture that, Frank. And picture upon one of those worlds inhabitants of proportionate smallness. What would they see, feel or think of the universe? Would they not conceive it about as we do? Picture them with powerful microscopes, looking downward into the matter composing their world. They would be aware of molecules, atoms—they would gaze down into Space unending. Another realm within their own. And within that one—others and yet others to infinity. The conception confuses you, Frank? It need not. Each of those realms is tiny—or large—according to the viewpoint. There can be no such thing as absolute size."

"That is what I mean," Dr. Gryce interrupted eagerly. "Absolute size—how can you conceive it? You can not. A thing is large or small only in relation to something else smaller or larger."

He waved his hand to the rolling landscape with the morning light and shadow upon it, visible through the arbor.

"There is our everyday world, Frank. How big is it? You can not say. Millimeters, meters, kilometers, helans, light-years—those are only words with which we designate a comparison. Compared to what our microscopes show us, this world of ours is very large, but compared to the spaces between the stars—the stars themselves—it is very small. Try then to imagine its absolute size. You can not, because there is no such thing. A universe within what we call an atom—another realm within an atom of matter upon one of the worlds of *that* universe—is not an extraordinary state of smallness *until we compare it with ourselves*.

"And this world of ours. It is normal to us; of no absolute size whatever—neither large nor small—until we compare it to something else. But suppose we visualize larger realms? Suppose we say these planets, stars—all the starry universe within our ken and this visual space which contains them—suppose we imagine all that to be contained within the atom of a particle of matter of some comparatively still larger realm? At once our world and ourselves shrink into smallness. Where a moment ago we had seemed large, now we seem small. Yet that other gigantic world within which we are contained—if we could live in it our telescopes would show us still larger Space unending. We would feel tiny—and of actuality *we would be tiny*—contemplating Space and size so much larger."

"And there you have infinity of Space," Brett added, as his father paused. "Unending Space both smaller and larger than ourselves. We—everything of which we can be physically aware—represent no more than a single step in the ladder which has no bottom nor no top. You can not conceive an end in either direction. There is no such thing. Nor—as Father says—can you declare anything to be small or large considered by itself alone. This then is Space as we conceive it to be. Illimitable, unending—infinite Space."

The conception momentarily seemed wholly beyond my grasp. What I would have answered when for a moment Dr. Gryce and Brett paused I do not know, for from the house the approaching voices of Martt and Frannie reached us.

"You'll fall, I tell you! Frannie, give me that!"

"I won't."

"You'll trip over the wires and you'll fall and smash it!"

"I won't."

The sound of a crash. And Martt's voice, "There, I told you!"

They were upon us, wheeling the tray laden with breakfast; Martt, flushed, laughing. "Oh, hello, Frank—they didn't switch you wrong, did they? Frannie broke the heater coils—if the breakfast gets cold, don't blame me."

And Frannie, also flushed and laughing and a trifle rueful over the mishap. Dressed in a blue blouse and widely flaring, knee-length trousers, with her golden hair tossing on her shoulders. The picture of a little housewife, of early morning informality. I thought I had never seen her so beautiful.

III

"That, Frank, is our conception of the infinity of Space."

With breakfast finished Brett had resumed the discussion. We were all seated in the arbor. Martt and Frannie momentarily were quiet, seemingly keenly interested in the impression upon me which they anticipated would come from their father's disclosures.

Dr. Gryce said, "The idea of Time unending is indissolubly bound with the concept of infinite Space. You will realize, Frank, for some centuries it has been understood that Time and Space are inextricably blended. We think instinctively of Space as a tangible entity—of length, breadth and thickness. And of Time, as intangible. Such really is not the case. Space has three dimensions—but Time also has a dimension."

"Length," Martt put in. "It sounds like a play on words, but—"

"It isn't," Frannie finished for him. "I can't imagine anything clearer than that Time has length."

Dr. Gryce ignored them. "You must understand also that Time as we conceive it can not exist except as the measurement of a *length* between two events. And what is an event? It presupposes the existence of *Matter*, does it not? Matter thus is introduced into the universe. It also can not be independent of Time and Space. So long as anything material exists, there must be Space for it to exist in; and Time to mark the passing of its existence.

"Of our universe, then, we now have Matter, Time and Space. There is a fourth—shall I say, element? It also is interdependent with each of the other three. It is *Motion*. You know, of course, that there can be no such thing as absolute Motion."

"Or absolute Time," Frannie put in.

"That we will discuss later," Dr. Gryce said quickly, "since it is more intricate of conception. Absolute Motion is impossible and non-existent. We can say a thing moves fast or slowly, *only in relation to the movement of something else*. One word more. I want you to realize, Frank, how wholly dependent each of these factors is upon the other. *Matter*, for instance, is an entity persisting in Space and Time. *Motion* is the simultaneous change of the position of Matter in Space and Time. A thing was *here, then*; it is *there, now*. That is Motion. You see how you can not deal with one without involving the others?"

"Say, Father, why don't you tell him what we're going to do?" Martt demanded. "Frank, listen—tonight Brett and I——"

"But I'm going, too," Frannie declared.

"You're not!"

I saw again that look of fear in old Dr. Gryce's eyes. His children—the spirit of youth with its lust for adventure—they were eager and excited. But Dr. Gryce saw beyond that—saw the danger. . .

He said gravely, "There is no possibility of my making you understand the details, Frank, until we have gone into the matter thoroughly. But as Martt implies, you are no doubt impatient. I will tell you then, briefly, that for most of my life I have been delving into this subject—Matter, Space, Time and Motion illimitable. Longing to investigate this immense material universe which I believe exists. But we humans are fettered, Frank. Like an ant, living for a brief moment enchained with a cobweb to a twig and trying to envisage the earth."

His voice now was trembling with emotion. "I was satisfied to see with my own eyes some little part into infinity. I invented what we—my children and I—call the myrdoscope. I will explain it presently. Suffice it now to say that there are normally invisible rays, akin to light, crossing Space, and I have made them visible. We captured them—saw after a myriad trials unavailing, occasional vague glimpses of the beyond which came to us. It might have satisfied me, but three years ago, one night, Brett saw——"

He paused, looking at Brett. Martt and Frannie were breathless, with eyes fixed on me.

Brett said, and his voice had a queer, solemn hush to it, "I was looking through the myrdoscope. We had seen blurred, brief glimpses of a realm ——"

"Beyond the stars," Frannie breathed.

"Yes, beyond the stars. A realm seemingly of forest, or something growing. Silvery patches—you might imagine they were water, or light shining upon something that glistened. They were always haphazard, these glimpses. We caught them, not always from one direction—seemingly from everywhere. A realm encompassing—enclosing—our whole star-filled Space.

"With the labor of years, which you, Frank, will appreciate to some degree, Father has charted what for our own little ken we might call absolute points in Space. Landmarks, say, of this outer realm. With our whirling earth, the ever-changing planets and stars, only this outer realm seemed of fixed position. We could sometimes return our gaze to the same landmark—a tremendous crescent-shaped patch of silver, for instance, which several times we succeeded in re-finding.

"It was near this patch at which I was one night gazing, when through some vagary of the ray bearing its image—or some difference in our crude apparatus—the scene suddenly clarified. And magnified as though at once I had leaped a million light-years toward it.

"I saw then a magnified section of the larger scene. The patch of silver appeared now as a shimmering, opalescent liquid. A segment of shore-front; and this all in a moment, again magnified. Upon a bluish bank of soft vegetation, with the opal liquid beside it, I saw a girl half reclining. A girl of human form, but transfigured by a beauty more than human. A girl of a civilization behind our own—or perhaps one in advance—I do not know. She was robed in a short, simple garment more like a glistening, glowing silver veil than a dress. Her hair was long—a tangled dark mass. She reclined there in an attitude of ease and the abandonment of maidenly solitude. I say that she was more than beautiful—oh, Frank——"

Brett's voice had suddenly lost the precise exactitude of the scientist. He seemed to have forgotten his father—Martt and Frannie; it was as though he were confiding his human emotions only to me.

"Beautiful, Frank. A strange, wild beauty, with a curious ethereal aspect to it. I don't know—it's indescribable. Human—half human, but half divine."

He checked himself; the scientist in him again became uppermost; but though he now spoke with careful phrasing, his face remained flushed.

"It was some moments before I saw additional details. And then I realized that the girl was not alone. Upon her bare feet were a sort of sandal with thongs crossing the ankle. And standing there beside one of her feet were two tiny human figures. In height, the length perhaps of her little foot. Men of human form; yet queerly grotesque; misshapen. One of them was in the act of reaching upward toward the tassel of her sandal cord where it dangled from her ankle; reaching as though to grasp it and draw himself upward.

The other was watching; and both were grinning with gnomelike malevolence.

"Nor was this all, for behind the girl, a brief distance away in what appeared a woodland dell, was another figure—a man of aspect akin to the grinning gnomes, save that in comparative size even to the girl he was gigantic. Ten times her height, perhaps, he stood behind her towering into the trees about him. A man of short, squat legs, dark with matted hair; a garment like the gnomes', which might have been an animal skin; a heavy massive chest; black hair long to his neck. A face with clipped hair upon it. He was regarding the girl; a grin, but with a leer to it—horribly sinister. And in his great hands, brandished like a bludgeon, was an uprooted tree.

"Have I given you an idea of motion in the scene? There was none. The girl was obviously wholly unaware that she was not alone. She lay motionless. But the lack of movement in her—in them all—was more marked than that. The girl's lips were parted in a half-smile of revery; but the outlines of her bosom beneath the silver veil did not move. There was no movement of breath; no change of expression. The gnomes, the giant—not the minutest change could I see mirrored in their faces.

"Yet it was so lifelike, I could not doubt it was life—and that the motion was there though I could not see it. I watched all night, shaken with this fragment of drama, perhaps tragedy, which I was witnessing—but even the girl's eyelids did not tremble. Dawn came; the scene faded.

"For a month I did not even tell Father; and Frank, the vision of that girl has never left me. The menace—gruesome, sinister—upon her—and her beauty ——"

"Haven't you ever seen her again?" I asked eagerly. "Was it life? How could it be life without motion?"

"Oh, he saw her again," Martt exclaimed. "I've seen her—we've all seen her."

"Tell him, Brett," Frannie urged.

"A month before I even told Father. During it, I searched for the scene unavailing, then Father and I searched together. It was a year, when almost from the same orbital position we came upon the scene again. A year—and now we saw a change. The figures all were there, frozen into immobility as before. But the gnome had caught the tassel, had drawn himself partly up to stand upon the girl's white ankle. The giant had come a trifle forward, and the upraised tree in his hands was partly lowered. The girl's attitude was unchanged, but there was now upon her face the vague dawn of startled knowledge, as though at that instant she was becoming aware of something pulling at her sandal cord, something touching her ankle—perhaps too, she was hearing a sound from the giant behind her. The startled knowledge which as yet had not had time fully to register upon her face."

My mind was whirling with a confusion of thoughts; the vague comprehension of what Brett meant was coming to me. I stammered, "Not yet had time—but Brett, you must have watched them all that night——"

"That night, Frank. And others—but there was no sign of movement. Another year—that was last year—we saw the girl partly aware of her danger. This year—a month ago—she was fully aware of it. Frightened—her eyes stricken wide with terror. But she had had no time as yet to move.

"Don't you understand, Frank? That drama is going on out there now. Like size of Matter and Space—and rate of Motion—there is no absolute Time. It is all comparative. To that realm out there of which we have been given a little vision, our tiny worlds here in the heavens are mere whirling electrons, like the electrons within one of our own atoms which to our consciousness of Time revolve many times a second.

"A year! A single revolution of our earth about its sun! To that girl out there, what we call a year is merely an electron in a fraction of a second revolving about its fellow. Even that is very slow—for she herself is wholly within the atom of a greater world outside her. A year as we call it—a second or less, to her. And though she is in full movement, how can we hope to see it by watching for a night? If a year were a second to her—an eight-hour vigil of ours would encompass less than a thousandth part of a second of her life!

"All comparative, Frank. There is nothing wonderful or really strange about it. In what we would experience to be a hundred years from now that girl will be fully faced with the menace of her assailants. A moment only, to her consciousness. It is that, Frank, we meant by the infinity of Time."

"Tell him what we're going to do," Martt insisted breathlessly.

It came from Brett in a burst almost incoherent. "I was not satisfied merely to see into this comparative infinity. Nor was Father. We have worked three feverish years, Frank, to climax all the labor of Father's which had gone before. And we have found a way—not merely to see, but to transport ourselves into these greater realms. A vehicle—I'll show you—explain it all. Its size can be changed—the state of the matter composing it is within our control. Its position in Space can be changed—simple enough, Frank, to enlarge upon the principles of our interplanetary vehicles. And—with one factor so interdependent upon the other—we have been able to control the rate of its Time-progress. It travels through Time as it does through Space."

His words were tumbling over each other. "You'll see it in a moment, Frank—test it—we have it here, ready yesterday. It sets us free, don't you understand? Free at last in Space and Time. And I'm going in it tonight—with Martt perhaps—we're going out to reach that girl upon an equality of Size and Time-progress. Going out to explore infinity!"

CHAPTER 2

"THIS COULD DESTROY THE UNIVERSE"

I had anticipated that they would show me a vehicle similar perhaps to the huge and elaborate space-flyers in the service of our Interplanetary Postal Division. But instead of taking me to the workshops where I had conceived it to be lying—serene, glistening with newness, intricate with what devices for its changing of size and Time-rate I could not imagine—instead of this they took me into the house. And there, in Dr. Gryce's quiet study with its sober, luxurious furnishings and his library of cylinders ranged in orderly array about the walls, I saw not one but four machines—mere models standing there on the polished table-top. Four of them identical—all of a milk-white metal.

But they were models complete in every detail. I stood beside one, regarding it with a breathless, absorbed interest as Dr. Gryce commented upon it. A cube of about the length of my forearm in its three equal dimensions, with a cone-shaped tower on top—a little tower not much longer than my longest finger. The cube itself had a rectangular doorway, and in each face two banks of windows. The door slid sidewise, the windows were of a transparent material, like glass. Midway about the cube ran a tiny balcony at the second-story level. It was wholly enclosed by the glasslike material. It extended around all four sides; small doors from it gave access to the cube's interior. The cone on top also had windows, and its entire apex was transparent.

I bent down and peered into the lower doorway. Tiny rooms were there. Bedrooms; a cookery—a house complete, save that it was wholly unfurnished. The largest room on the lower story—its floor had a circular transparent pane in it—was fitted with a seemingly intricate array of tiny mechanisms all of the same milk-white metal. A metallic table held most of them; and I could see wires fine as cobwebs connecting them. And in a corner of this room, a metallic spiral stairway leading to the upper story.

Dr. Gryce said, "That is the instrument room, complete. It contains every mechanism for the operation of the vehicle. We made it in this size—large enough to facilitate construction, but it is small enough to be economical of material. This substance—we have never named it—is of our own isolation. It is expensive. I'll explain it presently. . . . That room beside the instrument room is where we will put the usual everyday instruments necessary to the journey. Oxygen tanks—the apparatus for air purification and air renewal; telescopes, microscopes—my myrdoscope—all that sort of thing we can best obtain in its normal size. Those—and the furnishings—the provisions—all those in their normal size we will put into it later."

"You mean," I asked, "this is not a model? This is the actual vehicle?"

"Yes," he smiled.

"But there are four of them."

"We made six, Frank. It was advisable, and not unduly difficult to duplicate the parts in the making. The assembling took time——"

Brett said, "Father was insistent that we make every advance test possible. We have already used two of them. We are going to test the others today."

"Now," exclaimed Frannie. "Do it now—Frank will want to see it."

Dr. Gryce lifted one of the vehicles. In his hand it seemed light as alemite. He placed it on a taboret and we sat grouped around it.

"I shall send it into Time," he said quietly, "with its size unchanged, with no motion in Space, so that always in relation to us it will remain right here—I am going to send it back into other ages of Time." He turned to me earnestly. "We wanted you here, Frank, because you are so good a friend to me and my children. But for a selfish reason as well. When Brett goes out into Space and Time tonight, I want your keen eye to follow him. Your ability to record so accurately on the clocks what you see at any given instant——"

He was referring to my experience at the Table Mountain observatory—my first work when my training period was over. I had, indeed, a curiously keen vision for astronomical observation, and a quickness of finger upon the clock to record what I saw. In transit work I was extremely accurate; even now they were asking the Postal Division for my services at Table Mountain in the forthcoming transit of Venus.

Dr. Gryce was saying, "Your accuracy is phenomenal, Frank—your figures as you observe what little we see of this flight will help me—set my mind at rest that Brett is making no errors." He ended with a smile, "So you realize we have a selfish motive in wanting you."

"I'm very glad," I responded. He nodded and went back at once to what he had been saying previously. "I'm going to send this into Time. You must understand, Frank, that I can give you now only the fundamental concepts underlying this apparatus. We have so much to do today—so little time for theory. I need only tell you that it is readily demonstrable that Time is one of the inherent factors governing the *state of Matter*. This substance we have discovered—created, if you will—yields readily to a change of state. An electronic charge—a current akin to, but not identical with electricity—changes the state of this substance in several ways. A rapid duplication of the fundamental entities within its electrons—they are, as you perhaps know, mere *whirlpools of nothingness*—this rapid duplication adds size. The substance—with shape unaltered—grows larger. With such a size-change there comes a normal, correspondingly progressive change of Time-rate. We had to go beyond that, however, and secure an independent Time-rate, independently changeable, so that the vehicle might remain quiescent in size and still change its Time. In doing that, the *state of the matter* as our senses perceive it is completely altered. As you know, no two bodies can occupy the same space at the same time. Which only means that with the Time-dimensions identical, different dimensions of Space are needed. With the Time-dimension differing—the state of Matter is different; two bodies thus can be together in the same space."

"What is a Time-dimension?" I asked. "I mean—how can you alter it?"

"I would say, Frank, that the Time-dimension of a material body is the *length* —or a measure of the length—of its fundamental vibration. Basically there is no real substance as we conceive it—for all Matter is mere vibration. Let us delve into substance. We find Matter consists of molecules vibrating in Space. Molecules are composed of atoms vibrating in Space. Within the atoms are electrons, revolving in Space. The electrons are without substance, merely vibrations electrically negative in character. The nucleus—once termed proton—is all then that we have left of substance. What is it? A mere vortex—an electrical vortex of nothingness!

"You see, Frank, there is no real substance existing. It is all vibration. Motion, in other words. Of what? That we do not know. Call it a motion of disembodied electrical energy. Perhaps it is something akin to that. But from it, our substantial, tangible, material universe is built. All dependent upon its vibratory rate. And the measure of that I would call the Time-dimension. When we alter that—when through the impulse of a current of vibration we attack that fundamental vortex to make it whirl at greater or lesser rate—then we, in effect, have changed the Time-dimension."

There was so much that seemed dimly close to my understanding, and yet eluded me!

"But," I said, "if you send that little cube back into Time, it will no longer exist at all. It will be in the past—non-existent now. Or suppose you send it into the future? It *will exist* sometime—but now, it will be non-existent."

"Ah, that's where you're wrong," Brett exclaimed. "Don't you realize that you're making Time absolute? You're taking yourself and this present instant as fixed points of Space and Time—the standards beyond which nothing else can exist. That's fatuous. Frank, look here, it's simple enough once you grasp it. Time and Space are quite similar, except that you have never moved about in Time but you have in Space. Suppose you had not. Suppose—with your present power of thought—you were this house. You had always been here—always would be here. Suppose, too, that the world—the land and water—moved slowly past you, at an unalterable rate. That's what Time does to us. Then suppose I were to say to you—you as the house—'Let us go now to Great-London.' That would puzzle you. You would say, 'Great-London was here a year ago. But now it is gone—non-existent. It did exist—but now it doesn't.' Or you would say, 'The shore of the Great-Pacific Ocean will be here next year.' If I said, 'I'm going there now,' you would reply, 'But you'll be in the future. You'll be non-existent!' Making yourself the standard of everything. Don't you see how fatuous that is?"

I did not answer. It was so strange a mode of thought; it made me feel so insignificant, so enslaved by the fetters of my human senses. And these fetters Brett was very soon to cast off.

II

Martt said, "Can't we make the tests, Father? There is a frightful lot to do and it's nearly mid-morning already."

From the table Dr. Gryce took a small rod of the milk-white metal—a rod half a meter long and the diameter of my smallest finger. He knelt on the floor beside the taboret, peering into the tiny doorway of the mechanism he was about to send winging into the distant ages of our Past. Again we were breathless.

"More light, Frannie," he said. "I can not see inside here." Frannie illumined the tubes along the ceiling; the room was flooded with their soft, blue-white light.

"That's better." Rod in hand he turned momentarily to me. "I'm going to throw the Time-switch by pressing it with this rod," he explained. "Within the vehicle—the confined space there—the current is equally felt." He smiled gravely. "Without the rod I should lose a finger to the Past——"

Carefully he inserted the rod into the doorway. A moment of fumbling, then I heard a click. The little milk-white model seemed to tremble. It glowed; from it there came a soft, infinitely small humming sound. It glowed, melted into translucency—transparency. For an instant I had a vague sense that a spectral wraith of it was still before me. Then with a blink of my eyelids I realized that it was gone. The taboret was empty. Beside it, Dr. Gryce knelt with the rod melted off midway of its length in his hand.

I breathed again. Brett said softly, "It is gone, Frank. Gone into the Past, relative to our consciousness of Time. Gone from our senses—yet it is here—occupying the same Space it did before—but with a different Time."

He passed his hand through the apparent vacancy above the taboret. To me then came a realization of how crowded all Space must be! Of what a tiny fraction of things existent—of events occurring—are we conscious! That Space over the taboret—empty to me. . . . yet it held for a mind omniscient an infinity of things strewn through the ages of the Past and Future. What multiplicity of events—unseen by me—Time was holding separate in that crowded Space above the taboret!

Dr. Gryce was saying, "Let us test one now by sending it into smallness—come here, Frank."

He had risen to stand by the table, with another of the models before him. "This bit of stone," he said. "Let us send it into that."

He laid a flat piece of black-gray, smoothly polished stone on the table near the model. And with another rod he reached into the doorway. Again I heard a click. He withdrew the rod. "You see, Frank."

I saw that the rod was slightly compressed along the length he had inserted. The model was already dwindling. Soundlessly, untremblingly—it was contracting, becoming smaller, with shape and aspect otherwise unchanged. Soon it was the size of my fist. Dr. Gryce picked it up, rested it upon his opened hand. But in a moment it was no more than a tiny cube rocking in the movement of his palm. He gripped it gingerly with thumb and forefinger and set it on the polished black slab of stone. Its milk-white color there showed it clearly. But it was very small—smaller than the thumb-nail of my little finger. The cone-shaped tower was a needle-point.

A breathless moment passed. It was now no more than a white speck upon the black stone surface.

Brett said, "Try the microscope, Frank. You watch it."

I put the low-powered instrument over it; Brett adjusted the light. The stone was smoothly polished. But now, under the glass, upon a shaggy mass of uneven rock surface I saw the vehicle visually as large as it had been originally. But it was dwindling progressively faster. Soon it lay tilted sidewise upon a slope of the rock; smaller—a tiny speck clinging there.

"Can you still see it?" Brett murmured.

"Yes—no—now it is gone." The rock seemed empty. Somewhere down in there the little mechanism lay dwindling. Forever it would grow smaller. Dwindling into an infinity of smallness; but always to be with things of its size—and things yet smaller. . . .

As I turned from the glass, I became aware that Martt and Frannie were not in the room. Dr. Gryce and Brett, absorbed in the test, quite evidently had not noticed them leave. There had been two other models on the table—there was now but one.

Then from the garden outside the house a cry reached us. A shout—a cry of fear—terror. Martt's voice.

"Father! Brett! Help us! Help! Quick!"

We rushed from the room.

Crowning wonder, yet horrible! A surge of fear swept me. In the garden quite near the house stood the other model. Small no longer. It had grown—*was growing*—until already it was as large as the house itself. Around it the flowers, shrubs, even a tree had been pushed and trampled by its expanding bulk. It stood gleaming white in the sunlight, motionless save for that steady, increasingly rapid growth. Its windows and doors loomed large dark rectangles; its balcony was broad as a corridor; its cone tower was already reared higher than the nearest trees.

"Father! Help!"

At the doorway of the vehicle, standing just outside it, were the terror-stricken Martt and Frannie. They were holding the end of a long metallic pole which projected into the doorway. Struggling with its weight, striving to throw the switch inside.

We reached them. The expanding bulk of the gleaming side of the vehicle had pushed them back into a thicket of shrubbery. Near them a tree, uprooted as though it were a straw sticking upright in sand, was pushed aside and fell with a crash.

Martt and Frannie were livid with terror; breathless, almost exhausted with their futile efforts.

Martt panted, "We can't—lift the pole! It's—too heavy—too large inside."

Within the huge doorway, by the sunlight streaming through the windows, I could see the interior half of the pole, bloated by growth, huge, heavy.

Brett shoved Frannie away. "Frank! Here—take hold with us."

Dr. Gryce was with us. Together we four men got the interior end of the pole upon the table inside. A tremendous switch lever was there. But the pole slipped, rolled down. I expected it to break at the doorway point where it was so small outside, but it did not. The expanding doorway had pushed us farther back. Another tree on the other side fell. Above us the vehicle's tower loomed like a cathedral spire. Tremendous now, the vehicle had grown until it was almost touching the house. A fence had been trampled, had vanished beneath its giant bulk.

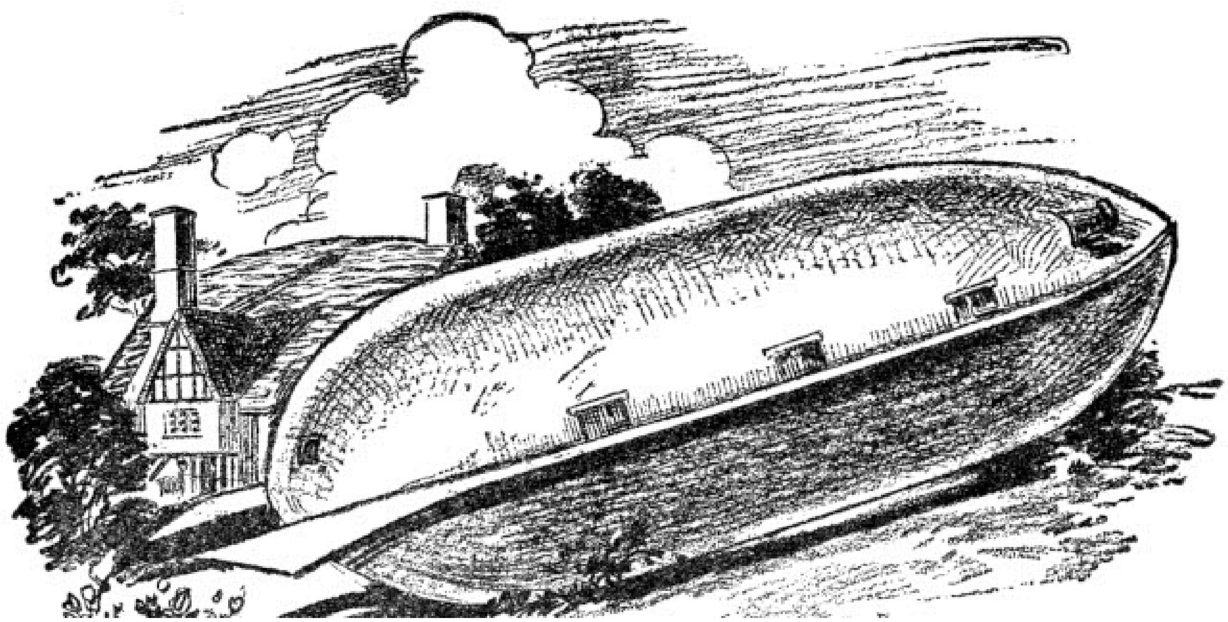
And the growth was increasingly rapid. If we could not check it . . . If it got wholly beyond control—this monster, growing . . . forever growing, to a size infinitely large—larger than our earth itself. . . .

I must have been standing stupidly confused. I heard Dr. Gryce imploring, "Take hold of it, Frank! We must lift it. We must—our last chance——"

But Brett pushed us away. "I'm going inside. I can move the switch—let go of me, Father! That switch—it isn't too big yet—but it will be in a minute. Let go of me!"

"No! No, Brett! The shock as you went in—you couldn't take it so suddenly. It might hurt you—kill you. And the switch is too big for your strength."

It was out of control—this monster, growing, inexorably growing—it was pushing at the house—a great white giant pushing gently but with an irresistible power at the little toy house beside it. I could see the house shifting on its foundations; a corner of it tilted downward.



"The vehicle was out of control, pushing at the house like a great white giant."

"Brett! Father! Try it now. One last try." Martt and Frannie had the pole again in position. With a last despairing effort we raised it; slid it up over the giant table-edge; caught the wide flaring side of the giant switch. Pushing—despairingly; five of us, pigmies struggling there at that giant threshold. The switch moved. Our pole held its place; the switch moved farther, clicked with a tremendous snap that reverberated about us. The growth of the monster was checked. It stood there serene, triumphant, with the little house, tilted, but still standing bravely beside it.

White, shaken, we ceased our efforts. Frannie gasped, "We—we only wanted to make it a normal size—so you could load it up with the furniture and things. But it—it got away from us."

Dr. Gryce said, "It is a lesson—perhaps a lesson which we needed forced upon us." He gestured to the great quiescent white building which had spread itself over most of the devastated garden. "A lesson," he repeated. "We must guard this power carefully. In unskilled or unscrupulous hands it is a power for evil almost unthinkable. This monster here—if it had gotten beyond us—if we had lost its control—this could destroy the Universe!"

CHAPTER 3

EXPLORERS INTO INFINITY

"You think we've got everything in it?" Frannie asked anxiously.

We had gotten the vehicle back to a size normal to our own stature; and all day had been working to equip it. The instrument room—its Space and Time and size mechanisms were complete. I had learned now that it was to be transported through Space by very similar principles to those commonly in use—a controlled attraction or repulsion of the faces of its cube for the heavenly body nearest to it; in effect, an intensification—a neutralization—or reversal at will of the electronic force which flows between and mutually attracts all material bodies; the force which once—in centuries past—was called gravitation. It needed no word of explanation. Its velocity and distance dials, its direction indicators, were familiar, though rather more intricate than those I had seen in the Interplanetary Service. Beyond that, there was a bank of dials upon which a changing size was recorded—with the vehicle's present starting dimensions to be the standard unit. And other dials for its Time-change. Of these there were two distinct sets. One, a record of the normal Time-change, inevitable to a change of size; another, a comparison of that Time-distance with the normal Time-progress of the earth, so that the Time-position of the vehicle into the earth's Past or Future could be seen.

In a subsidiary instrument-room was a variety of modern astronomical apparatus; the myrdoscope, and a receiver for an aural ray which, as a guide to Brett, Dr. Gryce was to send from earth. Of this, in more detail, they later explained.

In a smaller room were the apparatus for air renewal, the making of various necessary gases, water and synthetic foods; a store-room of provisions; rooms furnished comfortably so that the vehicle was complete in its living quarters. A thousand details, until at the last I felt as Frannie did—wondering how we could have failed to overlook a score of things we had intended to do.

It was nightfall when we finished; and all that evening we spent checking up the equipment. Dr. Gryce's home had not been seriously damaged by the morning's mishap; and as midnight approached we gathered in the little observation and instrument room he had built in its upper story. Brett and Martt, it had been decided, were to make the journey; we others were to watch and wait. It seemed the more difficult role. All that evening Dr. Gryce had been increasingly silent, careworn of manner and aspect. And though Brett was excited in his mature, repressed fashion—and Martt frankly exuberant—I saw that little Frannie was solemn, perturbed as her father.

It was a soft, brilliant, cloudless night, with no moon to pale the gleaming stars. And at last every detail was settled, and the midnight hour we had set for departure was at hand. We went forth with them to the waiting vehicle. There was nothing more to say. They stood—Brett and Martt—in the opened doorway as we gathered about them.

"Well—good-bye, Father—good-bye, Frannie dear." Brett held her close; then released her, pushed her away. "Good-bye, Frank." His hand-clasp was warm and steady.

Martt was jocular, but now at the last I could hear a tremble to his voice. "When we get to that girl out there—well, I'm going to tell her how interested you all are in her." His laugh was high-pitched. "That is, if we can handle that giant."

"Good-bye, Brett. Good-bye, Martt."

Our words were so futile, so inadequate to the surge of feeling within us! The door slid closed upon them. The vehicle, not to change size until it was far into the realms of outer interstellar Space, beyond our crowding little planets—lifted gently, soared upward, slid away from us, a glistening white shape up there in the quiet starlight.

Gravely, silently, with what sinking of heart I could only imagine, Dr. Gryce stood regarding it. Beside me Frannie was crying softly.

Explorers into infinity! And they were gone, to encounter—what?

CHAPTER 4

THE WATCHERS

We spent the rest of that night in the little observation room on the upper story of Dr. Gryce's home; with him and Frannie beside me I sat watching the vehicle's flight through the electro-telescope. It was not a high-powered instrument, but it served. I could see the vehicle plainly as it passed through our atmosphere and out into Space. A tiny blob with darker rectangles of windows.

Dr. Gryce sat with instruments, charts and his computations before him. Occasionally he would ask me for the vehicle's position; and I would give him the points and clock the time with all the accuracy of which I was capable. He seemed solemn, perturbed no longer; the scientist in him was all-absorbing. He said once with satisfaction, "Brett is competent—the boy hasn't varied a hair from my directions."

I knew that he and Brett had picked up the image of the girl and her assailants within a month past; and that Brett had accurate calculations which he could follow until able to capture the image on his own instruments.

"How long will it take them to get there?" I asked. "When will they be back? You said within a few days. How long?" Dr. Gryce looked up from his work with a faint smile. "There's no answer to that, Frank. Without a change of their time it might take them to reach that realm out there a thousand years or a million years—the vehicle's maximum velocity we do not know—that they are to find out."

"A million years! And another million to come back!"

His smile broadened. "As we measure Time, yes. But they will change their Time-rate; the trip may seem to them only a few days."

"But," I persisted, "two million years of our Time! And we can not change our Time."

"No, Frank. But you speak thoughtlessly. Brett can return to any point in our Time he wishes. Not with exactitude—but, we hope, within a few days. They will return here—within that Time we have agreed."

Frannie's face was very solemn though she said nothing; and I knew then that she was wondering if her brothers would be able to keep their promise.

Dr. Gryce rose from his chair. "I must adjust the aural ray—Brett may need it."

He had already explained this ray. A device similar to the familiar aurometer by which the aural power of the earth is measured. He had perfected an instrument for projecting into Space the invisible aura of the earth—projecting it in a tiny, very intense beam. An instrument for visualizing its characteristic bands was in the vehicle. They hoped that the ray might reach out into distant, interstellar Space; a flash of it crossing the sky as our earth rotated. And, coming back, Brett would see it, recognize it. A guide, as he came back from beyond all the universes strewn there throughout the magnitude of Space. If it could reach out there—if he saw it. My heart sank at the thoughts, doubts, which rushed upon me.

Dr. Gryce set his aural projector, with its ray, invisible to the naked eye, flashing after the vehicle. Silently he returned to his seat.

"Can you see them? You can still see them, Frank?" Frannie turned to me with anxious face.

I could still see the vehicle. But faintly, for faster than any mail flyer it was winging its way outward. Mars—approaching its closest point to the earth now to bring a deluge of the Martian Mails—red Mars at midnight had been above us. The vehicle had gone that way; and now, visually beside the planet, they were sinking together in the western sky. The stars were paling with the coming dawn. The east flushed with it, and presently I could see the vehicle no longer.

And as I turned from my instrument, I heard Dr. Gryce. "Why Frannie, girl! You're worn out! Come, it's dawn—they've vanished."

Little Frannie had fallen asleep.

CHAPTER 5

THE RETURN

We did not sight the vehicle the next night; it had seemingly passed beyond range of my instrument. With the myrdoscope we hoped to catch it, but could not. The night following was overcast with clouds. But we remained awake; Dr. Gryce seemed to feel that his sons might be returning. It was pathetic to me, observing him quietly slipping away from us at intervals to wander among the wreckage of his garden, gazing anxiously upward.

A week and still they had not come. What Dr. Gryce said to my Director I do not know; but he told me the Director was satisfied to have me remain away until my present business was finished. I had determined as much for myself. Not all the Directors in the Service could have taken me away from here, with Brett and Martt unheard from.

Like a beacon day and night we made sure that our aural ray was flashing its beam. But would Brett see it?

Another week. Still no sign. Doubts, fears, terrors assailed us. Were we watching, waiting futilely for what would never come? The thought was in my mind—and I knew it was in the minds of Dr. Gryce and Frannie—but never once did we voice it. Had Brett and Martt, perhaps, returned to our Past? With mechanism impaired, had they landed here in what we now called the Past—landed to find a wilderness of roaming savages? Or to find this little Space we now called a house and garden, a barren icy waste with men no more than beasts upon it? Or landed here in our Future? Ourselves dead, gone and forgotten? A great city here on this spot, perchance, with strange people and strange ways and nothing remaining of the loved ones they sought? Or were they lost and wandering in Space? Out there among myriad starry Universes hopeless to find our infinitesimal Solar System? Or lost perhaps in Time, wandering through the eons searching for the little centuries, years, days that identified their goal?

Or, again, perhaps they had safely reached that outer realm? Perhaps, once there, something had happened to prevent their return? In what we now called the Present, perhaps they were out there, transfixed, just as to our vision that strange girl and her strange assailants were transfixed—stricken of motion, with a passing of Time to us insensible. Transfixed out there now, to take no more than a few breaths, to move a hand, no more, during all the span of our own tiny lives?

II

I was sitting early one evening near the monight hour, alone with Frannie in the observation room. Dr. Gryce, in the room adjoining, had fallen asleep, worn by repressed anxiety and his now almost day and night vigil. We were talking in half-whispers; and abruptly Frannie voiced the fear that possessed us all.

"Oh, Frank, can't you see them? Please, you must! Oh, I'm afraid they're never coming back. Never—coming back."

It sounded so horrible. "Hush, Frannie. You mustn't say things like that." I put my arm around her, and suddenly like a child she flung herself to me; sobbed, and clung to me.

"Hush, Frannie. Don't cry—please don't cry. I'll look again. I might see them now. I'll try to."

I drew away from her; went back to my instrument. I had in mind to try the myrdoscope, but all our efforts with it during the two weeks past had been unavailing. It was a calm, clear evening. A broadly crescent moon was falling into the west. Mars was well above the eastern horizon; through the electro-telescope I looked that way. My circular field was empty. Frannie was checking her sobs, interested with hope renewed.

"Don't you see them, Frank?"

"No—not yet—*Yes*! I see them! Frannie, I see them!"

From visually above the red planet, out of nothingness a huge shape suddenly materialized. It had not been there an instant before; it seemed for the space of a thought, a transparent ghost of the vehicle; solidifying until even before I had told Frannie, I was aware that I saw it there. The vehicle unmistakable.

"They've come, Frannie! I see them! Call your father. Dr. Gryce! They've come! They're safe!"

How my heart leaped to be able to say it! Frannie was calling; and Dr. Gryce, no more than half awake, repeating, "They've come? They're in sight? They're safe?"

This gentle old man, how full of thankfulness his heart must have been! He came stumbling into the room. "Where are they, Frank? You can see them, lad?"

I could see them indeed—plainly, for abruptly I realized that they were no farther than just beyond the earth's atmosphere. And I could see also the conventional vane flying at horizontal above the vehicle's tower to denote that all was well within. They had come. They were safe.

They landed in the garden. Like a wafting feather the vehicle floated down under Brett's skilled guidance. It was of a size seemingly identical with the one it had upon departure, but evidence of its trip was everywhere visible. Its gleaming milk-white color was dulled. Its sides were pitted and scarred—the metal burned. A lower corner seemed fused into a shapeless lump.

The door slid open as we crowded forward. My heart was pounding. A sudden, irrelevant thought leaped to me—a thought, hope, that they might have brought back with them that strangely beautiful girl they had gone to rescue. A thought abruptly, fiercely poignant—yet with it a consciousness of its whimsicality that I—Frank Elgon—who loved Frannie Gryce, should be possessed of such incongruous desire.

The door was open. Brett and Martt—queerly garbed to seem almost strangers—were crowding there, with no one else behind them. But already I had forgotten the girl. Frannie's glad cries of welcome rang out; and Dr. Gryce's tremulous greeting; and I heard my own voice, strangely calm, "Well! Brett—Martt—you got back safely, didn't you? I'm so glad—we're all so glad!"

CHAPTER 6

THE FLIGHT INTO TIME, SIZE AND SPACE

They seemed not tired, but undoubtedly they were hungry, famished; and before they would say a word of those strange things we knew they had to tell, they made us feed them. "Regular food," as Martt laughingly called it. "By the code! We've eaten for months weird things supposed to be edible. My digestion is ruined."

Months! They had been gone two weeks and two days into a realm where those little sixteen days were no more than a tiny fraction of a second! Yet they spoke of months! It was very strange.

"Frannie! *Don't* ask me that again." Martt affectionately tweaked her chin. "We found her, I tell you. Wait till we've had supper—you'll hear."

They ate with the relish of those long deprived of accustomed food; and as we sat with them, forbearing to ask the eager questions flooding us, again I had that impression of the strangeness which had come to them. It was not only their manner of dress, though that of itself was extraordinary. They wore shirts of a colored cloth with a high rolling collar in front, low and open in back. Short trousers that were queerly wide and flapping at the knee, stockings that seemed of a soft gray leather and long-pointed shoes of a material I could not name. Over the shirt a short jacket, wide-shouldered and with sleeves that puffed and flared; and a skirt to it at the waist which rolled upward. Their hats—which Frannie rescued from the vehicle—were solidly wooden of aspect, with low circular crowns and triangular stiff brims.

The garb seemed grotesque; yet they took it so as a matter of course when once we ceased our comments—and they were so easy in it, so unconscious of it—that abruptly I realized it was my own viewpoint that held the strangeness. Between them, also, there was a difference of aspect—a rationality to their characters. The colors of their garments materially differed. Brett's clothes were more sober—less vivid, less extreme. His shirt was a somber brown; Martt's was a glaring green. Martt's jacket had additional bangles fastened to its cloth, it rolled higher in the skirt; tassels depended from his elbows longer than those Brett wore. His jacket sleeves were fuller; his trousers flared more, and were a more brilliant hue. But I will say that when after a time I became in a measure accustomed to his looks, Martt was very handsome; and he carried himself with a sort of swinging, debonair grace and swagger wholly attractive.

They were strangers to us in their mode of dress; no one regarding them could have named a nation of earth or any of the habited planets from which they might have come. Yet the strangeness went deeper than their clothes. They seemed older. A vague aspect of command seemed upon them—especially did it envelop Brett, like an aura sensed but not seen. Martt's old jocularity was unchanged; no dignity, no reservation, no aloofness with us had been added to the new swagger. Yet beneath his laughter there seemed always a hidden solemnity. And then I saw it all—this subtle strangeness that clung to them—I saw it lurking in their eyes. Memories mirrored there; memories of things no man had seen and felt before. Eyes—and more especially Brett's eyes—which had seen, perhaps, too much.

II

It was Brett who began their narrative; began it with the slow, careful, precise phrasing of the scientist anxious to avoid error of memory; to be exact of every fact and detail. On his lap he held a book of notes, and another book of the many dial recordings. He consulted it.

"Our recorded time of starting was four minutes past midnight. Sixteen days ago, wasn't it, Father? Sixteen!"

He gave a queer laugh but did not comment upon his thoughts. "I had determined to start slowly. Martt would have rushed us, but I thought that caution was best until we were quite sure of the workings of these mechanisms new to us.

"I did not record our passing above the earth's atmosphere. But the vehicle was inordinately hot from the friction of our passage. Perhaps I took it too fast—at all events we did not bother with refrigeration since in Space we would so soon need the heaters. We sat sweltering at the main instrument table with the dials before us.

"I think, Father, that I followed your instructions carefully. The dials were all set and operating. The size-dials stood motionless at unit 1. Our relative Time-dials were motionless at the original unit of earth Time; and the earth dial-chronometers ticked off the passing of your seconds and minutes. On the Space-dials—when first I chanced to notice them—we had gone some 900 miles. Our velocity then had picked up to 1,500 miles an hour and was swiftly accelerating. The Time was 1 a.m.

"It is slow getting through the atmosphere, but now we were fairly on our way. As you suggested, Father, I was heading just a point off Mars where I could hold Jupiter and Saturn almost in a line ahead of us. They were all there visible through our floor window—we had turned over and were falling toward them. I was using a fraction only of the earth's repulsion, and holding steady with the selective attraction of Mars and the star-field behind it."

"We saw your aural ray," Martt put in. He was earnestly intent upon Brett's narrative. "We saw it—I saw it—through the spectrometer. The swing of it was apparent even at that near distance. And we saw the Martian Mail coming in—they landed in Eurasia that night, I suppose. Say, they move in a hurry, don't they? And stop in a hurry when they get down close."

Brett went on: "We were still within the lower cone of the earth's shadow. But presently we emerged and came into the sunlight. The brilliant blackness of Space; and the cold by now had penetrated so that very soon we were glad enough to use the heaters.

"You know the details of a Martian voyage, Father. And you, Frank? This was no different except that having no necessity of stopping I reached a greater velocity than they generally obtain. A forty-hour trip, isn't it, Frank?"

"There's nearly always one of the minimum-distance trips at about that," I answered. "But you had some sixty million miles for yours. That's a lot longer than a minimum distance."

He nodded. "Yes. We came abreast of Mars—I suppose about a million miles away. Our Space-dials showed about sixty-two million miles traveled. We had been gone from you thirty-nine hours. Our average velocity had been something over a million and a half miles an hour, and with steadily increasing acceleration had reached then nearly three million an hour.

"That was as quick a trip as you anticipated, Father? But even so, we found it irksome. We alternated at the instrument board. Martt prepared most of the meals—beyond that and sleeping there was little to do. Except to watch for asteroids; but the mails have reported the region through there remarkably free of them this season. We saw none inside the Martian orbit closer than a million miles, which to such a low velocity as ours held no danger."

Dr. Gryce asked, "The air purifiers, Brett? You had no trouble?"

"No. Or very little, except just at first with the chlorate of potassium. I was telling you about passing Mars. We saw it rising slowly past us—saw it through a side window. A huge crescent, the sunlight on half its disk, but even the unlighted portion was plainly outlined. Above us was the thin crescent earth, with the sun behind it. The tongues of flame in the sun's envelope were plainer than I had ever seen them. We were falling away from the earth and sun, into the inky blackness of Space with its blazing white stars.

"During all this first portion of the trip we were eager to get more quickly advanced. Beyond Neptune's orbit, with the Solar System once behind us, we would feel like explorers, even though Nogar—he holds the record, doesn't he?—went once 27,000 million miles out."

Dr. Gryce put in: "His record was 27,600 million miles from our sun. At nearly five million miles an hour, which was his maximum velocity obtainable, that trip for the full return passage consumed—I think the total time was 461 days."

Brett went on, "That was the record. But even to go a single light-year at that velocity would have taken Nogar around 84 years—just going out a little light-year of distance, to say nothing of getting back! And we had so many thousands of light-years to travel even to get beyond the stars. It seemed stupendous—impossible."

"Naturally," said Dr. Gryce. "Impossible, of course, had you held to that size." They were directing their explanations at me. I nodded. "But you didn't stay that size?" I suggested.

"No, of course not," said Brett. "But for a time, we did—I was cautious from Mars to Jupiter, Father. Nogar plunged right through the asteroid region there—plunged through at nearly his five million miles an hour velocity. I held down to three million. We kept a close watch, though Martt had a somewhat terrifying experience. Tell them, Martt."

Martt flushed a trifle. "It wasn't my fault—at least I didn't think so. At a velocity like that the space there between the orbits of Mars and Jupiter is horribly crowded. Brett was asleep. I sat by the instrument table staring down into the floor window at the black firmament into which we were dropping. You people take a voyage like this as a matter of course—but it was my first time off earth, and the beauty of it—of the heavens—well, I tell you it impressed me. The black firmament—those blazing constellations beneath us—the full moon of Jupiter every moment growing larger like a white round lamp down there.

"Well, anyway, perhaps, I was lost in thoughts of it—when leaping up out of the blackness came a great round silver disk. A hundred times the size of our full moon. Then a thousand. It was below me, but off to one side. It swept past, so close I could see its barren, rocky surface—a range of desolate gray mountains; and I could see, too, its rotation, like a ball tossed into the air slowly rotating. Before I could think to do anything—even to make a move—the asteroid went past, out of my field as I looked through the floor window. For a moment I saw it rising past a side window and then it was above us—gone completely beyond my sight in a moment or two. I want to tell you I was frightened—I called Brett down at once."

Brett laughed. "I found him white, shaking like a tower-trembler. If a collision had really threatened, he could have thrown the main Time-switch. Thrown us suddenly into the asteroid's past or future—I had told him that—but when the danger came, he never thought of it."

"I never did," Martt confessed.

"How close did the asteroid pass?" I asked. "I saw one once, on a Martian trip——"

"I suppose we passed it at a distance of some three thousand miles," Brett answered. "But at three million miles an hour we were traveling that distance in three or four seconds. It was a narrow escape. The asteroid's attraction had drawn us aside from our course—but I soon rectified that."

"I meant to explain about attraction a moment ago, Frank," Dr. Gryce interrupted. "The attraction of the vehicle on our planets is why Brett could not yet increase his size. Jupiter and Saturn were pulling the vehicle onward, and in direct proportion to the mass, of course, the vehicle was pulling at them. An infinitesimal pull—but had Brett increased its size materially—while still close to our planets—the vehicle would have been a seriously disturbing element. I did not want that. Indeed, with any great size-increase, the vehicle moving out there would have thrown our whole system into chaos."

Brett said, "I was careful to obey you, Father. We were safely beyond Saturn—and Uranus and Neptune were on the other side of the sun—before I even touched the size-switch. From the orbit of Mars to that of Jupiter there are some 334 million miles between the points we crossed. We were about 112 hours making the voyage. I kept us well away—some ten million miles. But the planet was a beautiful sight, assuming every phase from full to crescent as we passed. You have never been so close, Father? Nor you, Frank?"

"Nor I," spoke up Frannie. She said it in a whimsical fashion of pathos, as though to make us all realize that she had been neglected.

Brett laughed affectionately. "No, nor you, little sister. Well, it's a beautiful sight. You can see it similarly in the telescope, but somehow, at the same visual distance the naked eye shows it indefinably different. A beautiful silver disk with the broad dark bands upon it and the red spot glowing like a lantern in its lower hemisphere.

"Our velocity was slackened for a time as we passed Jupiter, since I had to lose its great attractive force and turn a neutral side to it. But once by it, with it blazing as a gigantic thin crescent above us, I used a full power of its repulsion. We gained velocity rapidly. With the region of minor planets passed I had no fear of using all the velocity we could obtain. I think Nogar was unskilful in the handling of his vehicle; at all events, before we reached the neighborhood of Saturn, we had attained a velocity of seven and a half million miles an hour. It was the greatest velocity we reached."

"But," I exclaimed, "but Brett, at seven and one-half million miles an hour, in your whole life-time—whether you changed your Time-rate or not, you would have to live those hours—in a whole life-time at that velocity you wouldn't get one-quarter of the distance even to the nearest star!"

"No," he agreed. "But I began using the size-change after we passed Saturn ——"

I interrupted again. "I've been wondering about that—I don't quite see——"

"I'll make it clear to you, Frank, in a moment," Dr. Gryce put in. "Go on, Brett."

"We were well past Saturn before I changed our size at all. Our average velocity along there was six million miles an hour—it was a run of about seventy-five hours. We would have been—even at our maximum of seven and one-half million miles an hour—more than another 240 hours getting past Neptune's orbit. It was too tedious. We determined, since Uranus and Neptune were in other parts of their orbits—far on the other side of our sun—I decided that once we were well beyond Saturn, I would start our increase of size. We were seventy million miles beyond Saturn, with nothing of importance ahead of us but the distant stars, when I determined to start the change. The space there was comparatively deserted—a few asteroids—sometimes we could go nearly an hour without even sighting one.

"With Martt beside me—we were both a little timid about it, naturally—I threw over the switch and started our growth."

He paused for the length of a breath. "It was extraordinary—all our experience of the voyage from that moment was extraordinary. I hardly know how to begin telling you. . . ."

III

Dr. Gryce interrupted. "Just a minute, Brett—I want to make absolutely clear to Frank the principles involved in this change of size in relation to velocity."

"May I ask a question first?" I hazarded.

"All you like," said Brett.

"I'm wondering why in your normal size you could attain no greater velocity than seven and one-half million miles an hour. Theoretically, you know, a freely falling body will accelerate to infinity. And with repulsion added—a body, not only falling, but being *pushed* downward——"

Frannie said, "Nogar found his approximate limit at five million——"

"Our limitations were similar to his," Martt put in.

"I know," I said. "I remember in the public newscasting they said——"

"We found the same conditions," Brett put in. "Our vehicle—any vehicle traveling in outer Space—is not strictly a freely falling body. For low velocities—the general voyaging from here to Mercury, Venus or Mars—Space may almost be considered a vacuum. But it is not a vacuum, as we know. The imponderable, widely separated atoms of the ether—to use the ancient word—begin to be a factor at velocities over three million miles an hour. The drag became increasingly noticeable——"

"And the heat of the friction warmed us up," Martt put in. "At six million miles an hour we were hot, let me tell you. Sweltering—even with the full refrigeration units going."

"That friction held us to seven and one-half million as our limit," Brett added. "Anything else, Frank?"

"Yes, I was wondering about our aural ray here. Could you still see it?"

"Oh yes. Our sun of the Solar System had dwindled—small, but white and brilliant. With the naked eye the little star which was our earth showed very faint but distinguishable. With the aurometer—even using its spreading field of vision so that it embraced all that portion of the sky—we could see your beam sweeping slowly across the field as the earth rotated."

"And the myrdoscope?" I suggested. "Hadn't you tried again to locate the image of that girl?" My heart thumped as I said it.

He nodded. "Beyond Jupiter, when the long hours of inactivity hung on us, I spent many of them searching ahead of us with the myrdoscope. At last I picked up the image of the girl—held it for a few moments."

"There was no change?" Dr. Gryce said eagerly.

"No. The little distance we had traveled made no change—in fact, my smaller instrument, Father, showed it rather less clearly."

"I mean no change in the girl's attitude," Dr. Gryce insisted. "No change in the attacking giant—or those grinning little dwarfs at the girl's ankle?"

"None. But she was aware of them. On her face was stark terror—as we had seen it from here, Father, a month before. I noticed that the giant's forward step had nearly been completed—and the climbing dwarf was holding tightly to her sandal cord."

Brett gazed at me inquiringly but I shook my head. "That's all I have to ask," I said. "Go ahead, Brett. You were telling us about how you started the size-change——"

Dr. Gryce put in. "I think you had best proceed, Brett. And then if there is anything Frank does not understand, we can stop and make it clear."

He nodded, but for a moment he hesitated. "I flung over the switch to start our growth," he said slowly. "It was the beginning of all those strangely weird experiences which followed now one upon the other. Frightening at first. . . ."

IV

He paused briefly, then went on: "Our first sensation was one of shock—a reeling of the senses. But it was not severe—it passed almost at once. We found ourselves clinging there to the instrument table. To me the room seemed swaying dizzily. My forehead was damp with cold moisture; a nausea possessed me. I was oppressed; the air of the room was heavy to breathe."

"The air was snapping with the current," said Martt. "I could see it, and feel it tingling against my face. And it was heavy to breathe, as Brett says."

Brett resumed: "But we felt better after a moment. I saw the change first on the dials. The pointer of the lowest unit dial of the size series was slowly but visibly moving. I watched as it crept from 1 to 2. We had doubled in size. I gazed about the room. It was unchanged; and now as my body rapidly adjusted itself to the new conditions, I began to feel almost normal. Except a queer whirring in my head, and the nausea which persisted for perhaps an hour, I felt no evidence of the growth. The room, the vehicle was untrembling. No slightest evidence within the vehicle of the size-change going on—except the creeping pointer of the lowest dial. It was moving faster; it had reached 10. The pointer of the dial beside it—registering in units of a hundred—now seemed stirring."

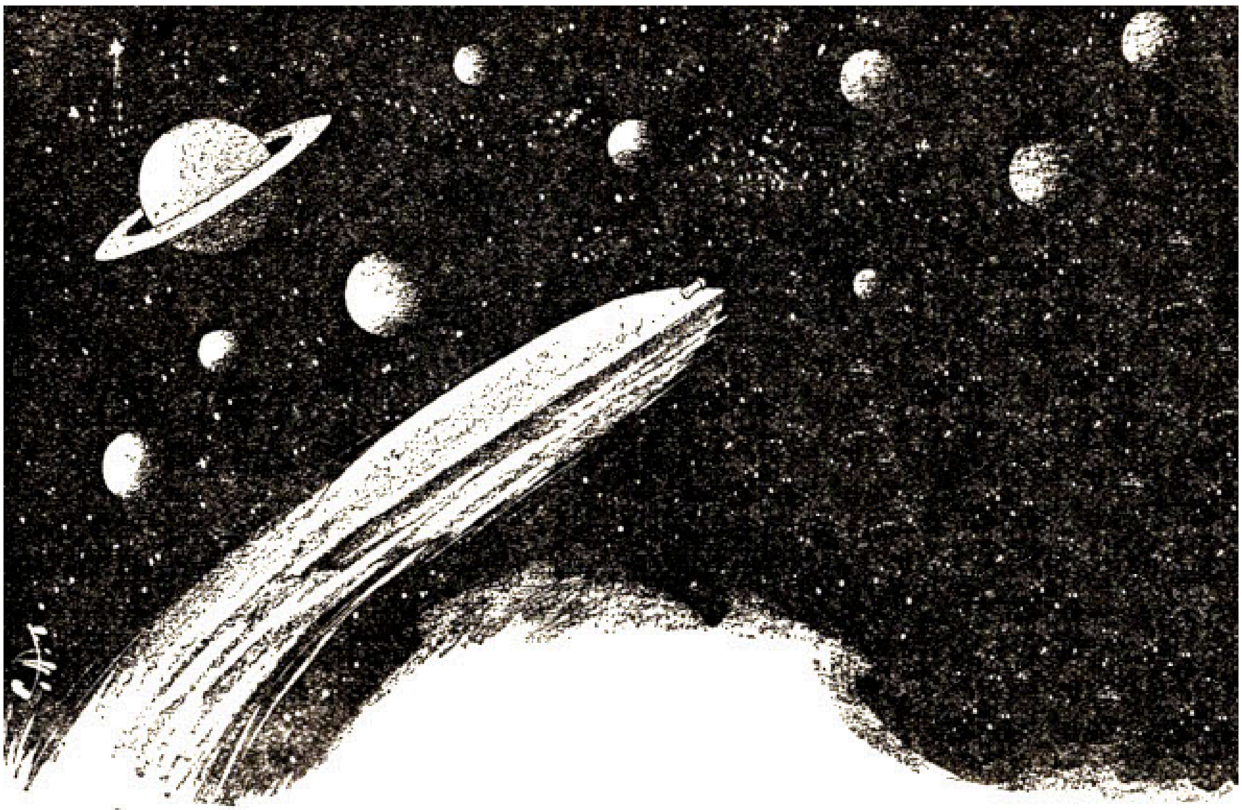
Brett gazed at us earnestly. "I want to make myself absolutely clear. We were then—I suppose a minute or so had elapsed—we were ten times our original size——"

"Much faster than the vehicle grew in the garden," I exclaimed.

"Yes. I had chanced the possibility of severe shock and thrown the lever at once to a quarter strength. Martt and Frannie, in the garden, had put it on only to the one-hundredth part of its power. At all intensities, the growth, you understand, constantly accelerates. At unit 10, which we reached in possibly the first minute, we were ten times our starting size—that is, for earth measurements, our vehicle from base to tower-top was then one-tenth of a mile. But soon the pointer had passed 50. And then 100—and the pointer of the hundred-unit dial had crept to 1.

"With recovered normality of senses we had gone to the windows. I want you to visualize first what always before we had seen. An inky black void everywhere surrounding us, in the center of which seemingly we hung motionless. The brilliant firmament of stars, freed from the distortion of earth's atmosphere; glittering, blazing like great diamonds. Pure white, blue-white, or tinged with yellow and red. The whole extent of the heavens swarming with them. The huge, spiral nebulous masses fleecy white, with tiny points of blazing white fire in them. And behind them all that distant ring of seeming star-dust—immeasurably distant yet glowing like a silver veil, which in the ancient books they called the 'Milky Way.'

"Near at hand, above us were the tiny planets of our Solar System. The sun, only a pale white disk from out here near Saturn; the earth—a star very faint; red Mars, a tiny reddish dot. But Jupiter was brilliant; and Saturn from our proximity was stupendously beautiful. The globe itself—a great silver disk, with the sunlight to make a narrow portion of it into a blazing crescent. The darkened areas of the globe, even on the shadowed portion, were plain almost as the bands of Jupiter. And Saturn's rings! Concentric rings—the inner one a trifle darker—opened up to a narrow angle—a glowing silver band like a broad hat-brim encircling the planet—a hat-brim over 37,000 miles broad.



"Saturn with its rings was stupendously beautiful from our proximity."

"This we saw, with ourselves of unchanging size. But now we were growing. The change was at first apparent only in the aspect of Saturn—since it was closest to us. The planet seemed to become a little smaller—shrinking and creeping toward us. A contraction of its size—and as though the space between us were diminishing. Yet—as a seeming paradox—the visual diameter of the globe and the rings remained almost the same.

"It is difficult to describe. We seemed moving closer to Saturn, yet in no sense was there any apparent motion. The effect—the result—of seeming motion—not the motion itself. Martt presently went back to watch the dials. He called out to me when we had reached unit 1,000. A thousand times our original size—the vehicle now ten miles in earthly height. The change had now affected very slightly the entire firmament. Everywhere a seeming contraction—not so much in the aspect of the blazing star-points, but in the black void of Space itself. As though the void were smaller—contracted so that everything in it were of necessity a little nearer to us. But it was as yet barely noticeable. I might even have thought it a psychological co-action with the change in Saturn's aspect—a change unmistakable.

"Saturn, as we grew, had been seemingly smaller and coming visually nearer to us. Yet our velocity away from it was—in our original size—seven and one-half million miles an hour. Can I make you realize that the effect of *both* motions was apparent? It was as though we were moving forward to lengthen a dwindling distance, with Saturn following after us simultaneously to shorten it.

"It was at the thousand unit point—ourselves then ten miles of earthly height—that I shut off the size-switch. Of visual diameter, Saturn had really not altered materially."

Brett stopped as though carefully to choose his words. "I'm striving to give you a clear picture. A distant object of great size may appear of the same diameter as something smaller and closer. But you can generally tell which is which. There is a difference of aspect—impossible to describe, but readily seen. Saturn was like that—the change in the planet was like a progressive change from the one condition to the other. It had appeared large and distant; it changed, to be smaller and closer. Just before I shut off the size-switch, when our rate of growth had become comparatively rapid, Saturn took on other motions—I'll tell you about them in a moment.

"Do I make myself clear? I want to. . . . With our growth checked, there was at once a striking, visual result. We seemed receding from Saturn so fast that its apparent diameter dwindled very rapidly—a normal dwindling of rapidly added distance. Presently it was a mere star—then a pin-point of light. Then it was vanished. Our other planets of the Solar System had preceded Saturn into invisibility. Then our sun itself became so faint a star that I lost it. We were beyond the Solar System—itself wholly lost to the naked eye among the great star-clusters enveloping it."

V

"Wait," I exclaimed. "There is so much I want to ask you, Brett."

Frannie interposed timidly: "Did you say, Brett, that on earth the vehicle then would have been ten miles in height?"

"Yes," he agreed.

She commented, "Then your relative Time-dials must have been visibly moving——"

Dr. Gryce hastily interrupted: "The practical workings of the inherent Time-change I want Brett to explain carefully. You did not move the vehicle in Time, did you, Brett?"

"No sir. Not then."

I must have looked puzzled, for Dr. Gryce added: "We mean, Frank, that the vehicle could have traveled in Time—in earth-Time, for instance, to go into our past or our future. Brett had not done that. But immediately the vehicle started a size-change, you understand, there automatically began a Time-change inherent to that growth. Normal to it, let me say."

"Oh, yes," I nodded. "I remember you explained that. In relation to its size ——"

"I'll put it this way," Dr. Gryce went on. "That girl out there is moving through Time at a definite rate. Let us say a year of our Time would be measured as a second of hers."

"Less than that," Martt interjected.

"Yes lad, I know. But those rough figures will serve for the present comparison." He turned back to me. "Keep that in mind, Frank. Now conceive Brett and Martt changing progressively upward in size, from what they are here on earth, to a size normal to that girl and the realm she lives in. A corresponding Time-change must take place. At every point of the voyage in Time and size, the relative values must agree; the vehicle's Time-rate always must be in inverse proportion to its position in size."

I nodded. "I think I understand. You mean that when in size the vehicle had progressed half-way from our size to the girl's, that then the vehicle's normal Time-rate would be half-way between our Time and hers?"

"Exactly, Frank."

"At this ten-mile size what percentage of the size-journey had been made?" I asked. I smiled. "I'm trying to imagine how large that girl may be."

Brett said quickly, "I'll tell you that later. It was some distance farther on before I could calculate such relative values even as approximations."

Frannie said, "At that point, Brett, the vehicle began speeding into the earth's future, didn't it?"

Dr. Gryce exclaimed: "Child, that will only lead us into philosophical discussion. Beyond the realm of mathematics——"

"I don't think so, Father," Brett said quietly. "I would say that since everything—Size, Time and Space—is relative, depending wholly on the viewpoint of the observer—that Frannie's question is simple enough. To me as observer—to my consciousness there in the vehicle—every given instant was the Present. The earth was out there in Space, revolving about its sun; rotating on its axis—its movements to my consciousness *faster* than before. To me it was the Present. The earth was there. I saw it through the electro-telescope. I also saw your aural ray through the aurometer. The ray swept the sky with a rapid sweep, since to my altered Time-rate the earth was rotating faster. But every given instant was my Present.

"However, compare my consciousness to yours on earth. The earth—rotating faster relative to me—had, while I watched there, made, let us say, a full rotation in that first five minutes of my vigil. Relative to me—it was the earth's future Time. I was gazing upon earth in its *tomorrow*. So I think that I was, as Frannie said, speeding into the earth's future."

Frannie was triumphant. Dr. Gryce said smilingly, "You put it clearly, Brett. But it's a philosophical and metaphysical viewpoint nevertheless. You spoke of Saturn's having another apparent motion near the end of your size-change?"

"Yes," said Brett. "As our Time-rate became materially slower, the speeding up of all the motions inherent to the planets grew visible. Saturn's rotation on its axis became readily visible through the telescope. And the globe began very slowly shifting sidewise—at nearly right angles to our course—the visual result of the intensification of its orbital movement. . . . You were going to ask a question, Frank, a moment ago?"

I had not forgotten it. "You were telling us, Brett, how you stopped your growth at the ten-mile size. Almost immediately, you said, Saturn receded into an invisibility of distance. The entire Solar System vanished into distance. You had been traveling only seven and one-half million miles an hour before changing size. It was the new velocity I wanted to ask about. The whole question of velocity relative to size."

"Relative!" Brett exclaimed. "That's the keynote to it, Frank. Two differing viewpoints, always. Keep them both in mind—the viewpoint of earth-size, and the viewpoint of the vehicle-size. I'll try and explain it now. Once clear to you, our whole experience will clarify to your understanding. Conceive, from your external viewpoint of earth, the vehicle out there in Space dropping with a velocity of seven and one-half million miles an hour. That was its maximum, owing to the ether-friction. It started to increase in size. Hence its mass grew—in proportion directly as the cube. As the mass grew greater, the atoms of the ether became of themselves relatively smaller, less ponderable, less capable of exerting their frictional drag.

"This should be very clear to you, Frank. In a vacuum, a feather and a bit of lead fall at equal rates. The mass—the weight—has nothing to do with it. But in air—where there is a friction—the heavier object falls faster. The vehicle was like that. Its mass, so enormously increased, gave it a greatly increased maximum velocity. It picked up velocity rapidly with its growth. The formulas involved are intricate—I need only say that after forty-nine minutes of traveling at the ten-mile size, we had again reached maximum. It was about 200 million miles a minute."

"A minute!" I exclaimed.

"Yes. That is 12,000 million miles an hour, as against seven and one-half million. The vehicle's length, breadth and width had each increased to a thousand times their former size. Its mass was the product of the three—hence one thousand million times greater.

"These are all approximate to the actual figures, you understand. Round numbers are less confusing. Our resultant velocity, however, was 200 million miles a minute, at the end of the first hour. We were well beyond the Solar System by then."

Frannie asked, "Brett, why didn't Saturn appear to recede until after you had stopped your growth?"

"That was merely optical, Frannie. Our velocity away from Saturn was steadily increasing. But with our increasing size, the space seemed dwindling—as though Saturn were following after us. With the growth checked there was a visual reaction—an apparent leaping away. It was merely optical. Anything else?"

"I'd like to know," I said, "the relation of your Time in the vehicle at the ten-mile size—its relation to our earth-Time."

"The proportion of one to one thousand," he answered readily. "Seven seconds to me, then, was about two hours on earth. Could I have seen the earth when I reached that maximum, it would have made a complete rotation on its axis—a day of yours—in a minute and twenty-four seconds to me.

"It's all clear, isn't it? Suppose I go back to the details of our trip? With ten miles of earthly size, at a velocity of 200 million miles a minute we were dropping into the black void of Space. The Solar System was lost presently, even to telescopic vision, but with the naked eye the firmament of stars was very little changed. I searched with the myrdoscope for the image of the girl, but did not chance to pick it up. We were hot again within the vehicle, from the ether friction—as hot as we had been before.

"Beneath us, in the star-field for which I was heading, was Alpha Centauri. It is, as you know, one of the very closest stars to our Solar System—to our earth. In miles, roughly some 25,000,000,000,000. Four and a third light-years of distance, 4.35 light-years to be exact. At 200 million miles a minute we would have been some eighty-eight days getting there."

"I couldn't have stood a trip so long," Martt exclaimed. "I told him we'd have to increase our size again. Nearly three months to get to the nearest star—with others a thousand times farther on!"

"There was no reason for us to stay so small," Brett agreed. "Out there, with the Solar System so far away, I had no fear of disturbing it."

Again I interrupted. "Brett, the vehicle's velocity was then much greater than the velocity of light——"

"About eighteen times greater."

"It seems inconceivable," I added. "Impossible for any tangible entity in Space to attain such velocity."

"Ah, but Frank, that's where you're using the wrong viewpoint," Dr. Gryce exclaimed warmly. "You're still imagining yourself an observer on earth. But take the viewpoint of the vehicle. Space was proportionately smaller than before. Brett gives you the earth-size figures in order to avoid confusion. From the vehicle's enlarged viewpoint, Brett, what was its comparative velocity?"

"About twelve million miles an hour," Brett said. "As against a former seven and one-half million. Not so great a change, Frank?"

"No," I admitted. "But——"

"But you can not quite grasp how the two velocities can be the same? Existing simultaneously in the same vehicle, only with a differing viewpoint?"

I think that was my trouble. I nodded, and he said at once, "To the larger viewpoint, Frank, the Space had diminished a thousand times, to make a thousand miles become as one mile. Not an *actual* change—a relative change only. But twelve million miles an hour, with distance diminished one thousand times, is the same as twelve thousand million miles an hour with the distance factor unaltered. You see that, of course. Or consider the relative Time-values. The vehicle's Time was seven seconds to about two hours. The exact figures were one to one thousand. In the vehicle we lived a thousand earth-seconds in one. Applied, then, to the two viewpoints of velocity, it gives identical results for the distance traveled. Whatever the factors involved—the earth-Time; the vehicle-Time; the Space relative to the vehicle; or to the earth; and the velocity, relative either to the vehicle-size or earth-size—the result must be mathematically the same. You see? And, Frank, in describing the progressive size-changes into which we now plunged, I shall give you always Space with earth-standards, and our velocity from the viewpoint of earth. It reached tremendous figures; but you are to remember always that of actuality they must be divided by the relative size factor. They were never greater than you would have expected the vehicle to obtain.

"I was saying that we were headed for Alpha Centauri. Again we started the growth. I threw the switch to its fullest intensity. Martt stayed to watch the dials; I sat on the floor, gazing down through the window at the star-field spread out beneath me. When my head had cleared from the shock of starting the growth, I sat absorbed in watching. Soon visible movements appeared. The star-drifts began to be apparent. And we were going toward these stars; the apparent shortening Space, added to our increasing relative velocity, made their approach visible. In the field to the sides of us, the stars were shifting upward. Those in front were spreading apart with a movement very slow but perceptible as we dropped toward them.

"I do not know how long I sat there; Martt occasionally would call to me from his post at the dials, but I hardly heard him. Alpha Centauri presently came rushing forward. As you know, it is a binary—twin stars a few hundred million miles apart, its components revolving about each other with a period of eighty-one years. It had been one blazing white point of light. Then it separated into two. They stayed visually small, for they were dwindling before the vehicle's growth; but they came rushing toward us. Soon I could see them separated by a narrow black ribbon of the void; and could see them revolving one about the other."

"An eighty-one-year period, and you could see it!" I exclaimed.

"Yes—a very slow movement, but I could see it. I would have passed between them—the ribbon of Space there was widening rapidly, the stars themselves had become great, blazing white-hot suns. But I was afraid of the heat; I altered our course to present a slightly repellent side. The firmament turned partly over. The two stars swung up past our side window; in visual diameter larger than our earthly sun—they mounted upward, closed in above us, drew together to form one; a sun at first; then a brilliant star; then faint, until with the naked eye I lost it.

"Beneath us, the star-field in front was rushing upward much faster now. The constellations opening; the stars shifting—everywhere was movement—strange movement, unnatural, fantastic. I confess, Father, that I was injudicious. Martt was absorbed, fascinated in watching the dials, and when occasionally he would call to me, I told him everything was all right."

"I didn't know what was going on," said Martt. "You told me to sit there and I sat there."

"Of course you didn't know what was going on," Brett smiled. "But I did, and I think for a time I lost my wits. The stars were thick and close around us. The nebulæ were opened into individual points of fire. Everywhere was movement, unreal. Stars rotating visibly; binaries shifting about each other; other stars shifting about each other; other stars seeming to enlarge in size, or to diminish, to swing this way or that with all the optical vagaries of our velocity, our changing Time and Size; and always those of the star-field in front—beneath us—spreading to the sides, rushing past our windows, closing in above us and fading into invisibility.

"A myriad universes in fantastic motion. And suddenly I realized that these giant suns were very close to us, and very small! Some I had recognized—blazing globes 100 million miles and more in diameter, and thought myself ten times that far from them. But it was not so. I stared at a giant globe 100 million miles in diameter, and with my viewpoint suddenly changed I saw that it was no more than a tiny glowing meteor, sweeping past a few miles away!

"All this star-field, little balls, rolling close upon us. A miracle that none hit us, though some time before, I had had the wit to call to Martt to make all the faces repellent. By inertia only, we plunged onward, repelling what lay in our path.

"I saw a wandering asteroid—a few hundred miles perhaps in diameter. It was whirling on its axis like a ball thrown into the air. A whimsical humor—a madness perhaps—had descended upon me. There was nothing but the asteroid momentarily close before us, and I called to Martt to throw attraction into the bottom of the vehicle. The asteroid came rushing. But shrinking—shrinking until I laughed aloud to see it dwindle to a ball I could have held in my hand; and dwindle further until impotently it struck the floor window with a tiny point of fire from its fusing rock and metal. A burning cinder which scarce would have hurt me had I caught it in my naked hands.

VI

"How long my mood of ironic madness may have lasted I can not say. I barely noticed our actual entry into the Galactic Plane. Enormous suns whirling past, now relatively not many times bigger than the vehicle itself. Others, distant a mile or so—or a billion miles if you want the other viewpoint—with their magnified drift making them dart crazily past. I gave no heed to passing time; I remember only that at last the star-field beneath us was thinning out. Stray clusters—a myriad glowing little balls hurled aside by our rush. But there were visibly less and less of them, until, quite suddenly, I realized that unbroken inky darkness lay ahead. And to the sides and above us, the star clusters, nebulæ swirling like silver mist—it was all fading. Winking little points up there behind us—winking and vanishing.

"We were in blackness unbroken. Dropping into a void of blackness with velocity inconceivable. Suddenly I was frightened. Stiff from so long upon the floor, I rose and hurried to Martt. We shut off the size-switch; made all the faces repellent. But there was nothing to repel; nothing to stop our downward rush into that blackness. It seemed all at once a blackness pregnant with unseen things of fearsome aspect. . . . The size-dials showed us to be near unit 50,000,000. Fifty million times our original size! The vehicle 500,000 miles high!

"The relative Time-dials—showing relative earth-Time—were whirling. Our Time in the vehicle was less than a single second to a year on earth. My mind leaped back to you. Every second we lived there in the vehicle you here on earth were living more than a year. A century of yours was little more than a minute to us. The earth's future, whirling on a thousand years while Martt and I sat there confused at the instrument table. A tiny little earth, spinning like a top upon its axis, flashing around its tiny sun with a complete revolution every second!

"The velocity indicators, as well, were in rapid motion. The indicator of the miles-per-hour unit was an indistinguishable blur. And miles per minute—and per second—we could read none of them, so fast were they moving. The light-year distance pointers were in motion. We were piling up light-years of distance every moment. The total stood—as momentarily I read it—at between eleven and twelve thousand light-years of total distance traveled. Light, speeding at 186,000 miles a second, must go a year to make a light-year unit of distance. And we had gone nearly 12,000 light-years! I read our present velocity on the light-year velocity-dial. It was 3480 light-years per hour! And still rapidly accelerating!

"The panic of fear possessed us at the strangeness of it all—at that void of blackness—soundlessness—into which we were plunging; and even our plunge unmarked by the faintest trembling of the vehicle. A panic. I started to use the aurometer to search for your ray. Absurd! The absurdity of it made me laugh hysterically. Your ray had been extinguished thousands of years in my Past. I tried the myrdoscope—to locate the image of the girl—to verify our direction, for abruptly I realized I had, in that empty black void, nothing by which I might locate our position.

"The myrdoscope was inoperative! I could not locate the girl-image—nor anything else. I tried with the electro-telescope at its greatest power—tried frantically to pick up some star-image behind us. I could not. I did not think they were as yet beyond its range—it merely had gone dead. The current in it would not hum. It was dead like the myrdoscope. We wondered then if our dials were working accurately. In our panic we doubted everything. And knew, with a stark terror upon us—knew that we were lost. Lost perhaps in Size and Time. And lost in black Space, empty, soundless, unfathomable!"

CHAPTER 7

"A SINGLE STARLIT NIGHT—AN ETERNITY"

Brett had momentarily paused in his narrative, but when we would have plied him with questions he waved us aside.

"Let us finish first. The panic that was upon us with this knowledge—belief—that we were lost out there in Time and Size and Space did not last long, for we fought against it. And presently we were calmer—able to reason. Our size-dials were at rest—we had shut off the switch. By earth standards the vehicle was 500,000 miles in height. Our relative Time was a century of yours, to a little more than a minute of ours. Some 8,000 years into your earth-future had already piled up on the earth standard Time-dial—and we were adding one hundred years to it almost every minute. Our velocity had reached a maximum of 3480 light-years per hour—and we were 12,000 light-years from earth. The velocity was now lessening a trifle; it dropped nearly to an even 3,000. With unchanging size now, with nothing near us to repel or attract, the ether-friction overcame inertia to reach a balance of forces.

"We conquered our fear—began to reason what we should do. It was of course futile to look for your aural ray. It had been extinguished thousands of years. We wanted to go on to our destination, and it was the non-operation of the myrdoscope which worried and puzzled us. . . . I was sure, Father, that up to this point in the voyage I had made no serious error of direction. The image of the girl should have been before us. But the myrdoscope would not work."

"The Time——" I suggested.

"Ah, no, Frank! We had progressed very little into the Time of that girl's life. She should still have been reclining there on the bank; or at least the bank itself should have been there. We puzzled over what could be the trouble with the myrdoscope. We found the trouble——"

"I found it," said Martt eagerly.

Brett nodded. "Yes, it was Martt who reasoned it out. A curious explanation—and one, I think, which involves the greatest of all the issues we had encountered. The myrdoscope would not operate for a very big, but very simple reason. You would think to find the answer in Science? Not so. It was a theosophical reason, Father."

Brett was very earnest, and very solemn. "It was my purpose, you understand, to reach the girl at the *exact moment* we had always seen her. We planned to make our Time before reaching her, coincident with hers of that given instant. Remember that. Consider then: At this other instant when now we were trying to see her through the myrdoscope, our Time-rate had carried us about 8,000 years into earth's future. But also, it had carried us some forty minutes into the girl's future.

"Not science now. Metaphysics, perhaps—and certainly Theology, and Theosophy. We were destined *to be with the girl during those forty minutes*. And we could not now look ahead and *see ourselves*—see our future actions.

"Father, you've spoken of that. What you said was true. It is not God's way that man should look at his own little future. Not best for us. The Almighty knows it, and has prohibited it. Chaos would result, for we live upon hope. There was no scientific reason why the myrdoscope should not show us what we were destined to do during those forty minutes. Yet—it was dead. Dark. Inoperative.

"And this now I know: With all the science in the world there are some things you can not do—those things which transgress the Creator's laws. Before them—against all scientific reason, logic—we must fail. You can not see your future; you can only live it once. Nor can you go back through Time to stop in your own Past; to live again your life—to do differently than you did before. It is unthinkable—impossible, even though now we have the scientific means to accomplish it. It is not the Almighty's plan—and He will not let us do it.

"We reasoned all this out. It was simple enough. We had our Time-switch which would change our Time-rate irrespective of the normal Time-change inherent to our size. . . . That was what puzzled you awhile ago, Frank? Well, now we used that Time-change mechanism.

"It brought us new sensations. A shock, a queer humming lightness pervading the vehicle, the air, our own bodies. A lightness as though almost we were mere shadows of our former selves. Specters, a ghostly vehicle, humming with an infinite vibration.

"Presently that all wore away; or at least we grew used to it—so that had there been anything in Space to see, as very soon there was, ourselves were the substance—all else the shadows.

"We went backward very slightly in Time. I suppose some forty minutes of the girl's Time. I tested it by the myrdoscope. The instrument flashed on! It was operating! A continuous *retrograde* action of the Time-mechanism was necessary to hold us upon that single given instant of the girl's existence. The calculation was intricate; I reached it, partly by mathematics, partly by experimentation with the myrdoscope. I saw fragments of the girl's immediate Past, as our Time-change swung us into it. Saw her arrive alone in the woodland dell. Saw her lie down, at ease, with a security unsuspecting; saw the grinning, vicious little gnomes creep upon her; the leering giant appear. And made, then, another startling discovery—I'll tell you about it in a moment.

"At last I had the Time-change correctly gaged; we were—in relation to the girl—standing still in Time. Presently we again increased our size. An alteration of the Time-mechanism was needed; a progressive alteration. But this was simple to calculate and to adjust."

Frannie asked, "What was your discovery?"

He smiled. "Curious as always, little sister? It was that the giant was in the act of becoming *smaller*! The gnomes were growing in size!" He checked our chorus of exclamations.

"I will tell you now: This giant—these gnomes—were three beings who did not belong to the girl's world. They had come there from a greater world outside the atom. By means of science—such means possibly as we now were using with the vehicle—they had diminished their stature to the infinitely small. Had gone down and down into their tiny atom, to come upon the girl and her realm."

II

Again Brett waved us aside. "Not now, please! Oh, yes—I can tell you the structure of this, our little fragment of the material universe! But let me finish first about our voyage.

"With our Time-change corrected, the myrdoscope readily had picked up the image of the girl. A larger image, for we were 12,000 light-years closer to her. The same scene, stricken again of motion. The giant standing there; the gnome climbing upon the girl's ankle; and herself, just aware of her danger, with dawning terror on her face.

"The electro-telescope also was working now. Looking behind us, we could just see the last of the stars. And soon they were gone. A day of our conscious existence went by. At 3,000 light-years an hour we added 72,000 light-years of distance—a total from earth of about 84,000. The black abyss of Space had not remained empty. Off to one side had been a faint glow. A nebula; a patch of star-dust. Through the telescope we could see stars—a complete starry universe. It was as large, no doubt, as that we had passed through.

"It gave us a new idea of the immensity of Space. Separated by some 30,000 light-years from our own universe of stars—of which the Solar System is so tiny a part—this other star-patch was equally as large. And yet it seemed to lie isolated in fathomless Space. It drifted by us and in a few hours was gone. And far off to the other side of us, another patch came past. And others; each several thousand light-years in extent; each isolated from all its fellows.

"We traveled another full day. Over 150,000 light-years from earth. Yet the girl's image was seemingly not coming nearer very rapidly. We felt the voyage would take too long, so again we increased our size."

I interrupted. "Had you calculated the girl's relative size?"

"Yes," he said. "In a moment, Frank, you shall have it. We—our vehicle—was 500,000 miles high, compared to earth. We increased it to 600,000. Our velocity also increased. At a million miles of height—I have made all my stated figures round numbers, but they are approximately correct—at this million-mile height, we reached normality to the girl. It simplified our mechanism adjustments. There was no longer a size-change necessary. A retrograde Time-change, equal to our own now normal rate of existence, held us at that same instant of her life.

"Our velocity was more than proportionately increased. To demonstrate that mathematically would be intricate—would involve several very complicated formulas, which would not interest you now. . . We passed, distantly, a score or more of starry universes—to the sides, and above and below us—lying in every plane; and of every size and general extent. Some were small, a few thousand light-years like our own. Others immense; one which seemed 500,000 light-years at least in diameter.

"We reached ultimately a maximum velocity of about 90,000 light-years an hour. We had previously gone 150,000 light-years from earth. We traveled some eighty additional hours, not all at the maximum—for possibly half that time we were steadily accelerating. And at a total of 4,750,000 light-years from the earth, a faint glow of seeming phosphorescence showed in the blackness beneath us.

"There was a universe to one side, ahead of us. But this was a different light. A radiation from the Inner Surface itself. The Inner Surface of the hollow little atom within which all this Space and its infinitesimal whirling electrons is contained. They are immense suns, to us here on earth, but from the larger viewpoint they were mere electrons, whirling, flashing around in tiny orbits a thousand times a second.

"The girl and her realm, as we had thought, are on this Inner Surface of what we may choose to call an atom. Themselves—this girl and her people—are infinitesimal. This atom of ours is merely some tiny particle of matter in that other world from which the giant and the gnomes had descended. A tiny particle of matter. Call it a grain of sand, lying with trillions of its fellows upon some great ocean beach—lying there in the light of stars shining in infinite Space above it. Lying there for a single starlit night which is all eternity for us. A single starlit night—an eternity! Infinity, of Space and Time? Why, even now I have seen no more than an infinitesimal fragment of them! . . . .

"The giant and gnomes were doubtless normally of the same size—only momentarily did they happen to be different. . . . Wait, Frannie, please! I can't tell it to you any faster. . . . The Inner Surface became visible to our telescopes at about 4,900,000 light-years. A realm of land and water.

Vegetation. Strange of aspect, yet normal too. It stretched beneath us in every direction—a huge concave surface.

"We kept our size, but using the repellent force of this Inner Surface, I gradually cut down our velocity. Down more and more until that last light-year or so took us a week to traverse. The girl, Father, is approximately 5,000,000 light-years from here. We—our earth—may be near the center of the void. I don't know. Perhaps we are much nearer the girl's side. It isn't important . . .

"The Inner Surface at last lay close beneath us. It took us an additional week of diminishing velocity to reach its atmosphere. I was cautious; I had the velocity under control always."

He paused a moment, seeming carefully to consider his next words. "I want you now to forget earth standards. Take the larger viewpoint exclusively. Let me speak of miles, not in relation to earth, but miles—in relation to the Inner Surface—which are 100 million times longer. Let me speak then of myself as again but six feet high; the vehicle, 52.8 feet high. Realize that by the larger standards I was but one-twentieth of a light-year from earth."

Dr. Gryce said gravely, "Your telescope would show a globe like the earth very plainly at one-twentieth of a light-year of distance. You must explain, Brett, why you could not see it—or any of the great stars of our immediate universe."

Brett nodded. "We could not see the earth, because to our size it was merely a little orange. To be more exact, a ball about five inches in diameter. A tiny ball I could have held in my hand, whirling out there in Space, spinning like a top on its axis to make your infinitesimal days and nights; traversing its entire orbit—a complete revolution around its little sun—more than three times every second!

"With these other standards, then, I want you to visualize us as we sat on the floor of the vehicle gazing down through the lower window. We were, say a hundred miles above the Inner Surface, just entering the upper strata of its atmosphere, and falling gently downward. Beneath us lay a broad vista of land and water; vegetation; forests; here and there patches of human habitation—houses, villages. It was a strange, unfamiliar landscape, yet not unduly abnormal. In every direction—as we dropped closer—it spread upward to our horizon. A rolling country; gently undulating hills, broad valleys—and off near the horizon a jagged mountain range. It seemed not far away; we could see black yawning holes in it; the mouths of caves, or tunnels, perhaps.

"The broad crescent lake lay directly beneath us. Trees bordered its banks; trees strange of shape—yet one would call them trees at once. A collection of low, flat-roofed buildings lay beside the water. A village—or a city. The buildings were queerly curved—seemingly crescent-shaped. They had no straight lines. They seemed generally of but one story, though a few were larger; and upon an eminence near the water stood one much larger; more ornate of shape than all the others.

"It was not a fantastic scene, but wholly rational to our own accepted standards. A sylvan atmosphere seemed to hang upon it. Trees and flowers were everywhere; the roof-tops seemed gardens as luxuriant as those beside the houses. The streets were broad and orderly; and beyond the city ribbons of roads wound out over the hills.

"A sylvan landscape, with an air of quiet peace upon it. I felt a sense of surprize. This was not modernity; nor a civilization more advanced than our own—nor yet was it barbarism. Later I knew it was decadence. A people who once had been far up the slope of civilization, over the peak, and now were coming down upon the other side. The peaceful, restful ease of decadence, which to complete the inevitable cycle of all human life ultimately would again bring them to barbarism.

"We saw these details as we fell gently toward the crescent lake. You will notice I have not mentioned color in the scheme, nor movement. Our Time-mechanism was operating. The scene beneath us was stricken motionless, since always we were holding to the same instant of its Time. An unreality lay upon it; a flat, shadowy grayness of aspect. An unnatural stillness. We dropped closer. A shadowy boat seemed on the lake—a boat with a sail. It lay there, immobile. The water was rippled by a breeze; but they were frozen ripples. And in the streets now we saw people and curious vehicles—all standing like waxen figures.

"The grove of trees—the woodland dell wherein the girl was lying—was a short distance down the lake shore from the city. A single house was near it; but in the other direction was unbroken forest. An open space was there—a few hundred feet from the girl and her assailants. We decided to land there. We knew we were invisible as yet—a ghost of a vehicle, all in this same instant coming from Space to land upon the lake shore.

"We had not yet decided just what we would do. But it was necessary to land first. And necessary also for the vehicle to assume the Time-rate of this realm before we could leave it. When that was done we would be normal humans, to rescue the girl as best we might.

"We dropped into the little clearing at the edge of the lake, and gently came to rest—and upon the surface of the ground, since to us it would have had no substance; but within a foot of it, where, like a ghost hovering, I held us level. The unreality of us, I must repeat, was not to us apparent; we seemed solid—it was the ground, the forest about us which was unreal. Spectral trees; a gray twilight. I made sure that nothing was touching us. We were a few inches only above a soft-looking gray ground. We were ready to cut off our Time-change—to take our places normal to this new realm."

CHAPTER 8

THE ENCOUNTER IN THE FOREST GLADE

Martt said, "I would have thrown off the Time-switch and rushed out at once. But Brett wanted to talk about it."

Brett smiled. "It was difficult for us to remember that no haste was needed. No haste—until we took the girl's Time-rate. And then we would need all haste possible. We discussed what we were to do. We had weapons—the electronic flash, for instance, with which we could have struck down that giant as with a lightning bolt. But could we? I was not sure—not absolutely sure—that the weapon would be operative. Or that, perchance, this giant would not by some strange means be proof against it. A man sixty feet tall is no mean adversary. Suppose he held the girl before him? Would I dare attack?"

"I suggested," Martt put in, "that we take the normal Time-rate of the girl, and be in hiding until the giant's size had dwindled to hers. The dwarfs were growing. But there would only be three of them, against two of us—and so far as we had seen, they were not armed."

Brett went on: "That didn't seem a good plan. The giant's size was, we had calculated, rapidly dwindling. Within five minutes he would be the girl's size. But suppose, instead of standing there during those five minutes he picked up the girl—made off with her? It was too dangerous.

"At last we decided to make the vehicle, and thus ourselves, somewhat larger. At the risk seriously of frightening the girl, we decided to take a stature larger than the giant. Thus, since he was not armed, we would have little difficulty keeping the girl from harm.

"The forest glade within which our vehicle was hovering was ample for the growth. We adjusted the mechanisms; and in a few moments of growth we had reached the determined point. We shut off the switches; the vehicle fell its few inches to the ground. . . .

"The scene clarified. We were in a somber forest of dull, orange-colored vegetation. Above us was a deep purple sky, with a few drifting clouds, and stars gleaming up there in the darkness. They were the stars of that last universe we had passed; unnatural of aspect, for they seemed unduly close and unduly small.

"It was not day—nor yet was it night. A queerly shimmering twilight; shadowless, for the light seemed inherent to everything.

"We were aware of all this in an instant, but we did not stop to regard it, for Time now was passing. The girl and her assailants were now, we knew, in full motion. With the flash cylinders in hand we stepped hastily from the vehicle doorway.

"The forest trees were saplings no higher than ourselves. We plunged through them, came to the other glade. The girl was sitting up with hands pressed to her breast in terror—a tiny figure of a girl not as long as my hand. The dwarfs were so small I did not see them at first; they were standing beside her—an inch perhaps in height. The giant, with what drug acting upon him we could only guess, had dwindled until he was only about half our own present height. He had dropped his tree-bludgeon, which now was too large for him, and was stooping down to seize the girl. His leer, with the reality of motion upon it, was horrible.

"Momentarily we had stopped at the edge of the glade. The figures there were aware of us. The girl screamed—a little voice, shrill with terror, an agony of sudden fear—at her assailants, and doubtless most of all at ourselves. The giant—I can no longer call him that, since we saw him as no more than three feet tall—at our appearance he straightened. Stared at us. Surprize, then fear swept his ugly hairy face. He shouted something to his tiny companions.

"The girl screamed—a little voice, shrill with terror, an agony of sudden fear."

"Martt's hand went up; he fired his cylinder. But he was confused—and the nearness of the girl to his mark made him aim high. The bolt missed; lodged harmlessly in a tree with a ripping of its bark. I rushed forward to seize our adversary, but he eluded me, leaped over the girl. I was afraid of trampling her—I stepped backward—clutched Martt, fearful of what he might do.

"It had all happened in a moment. The dwarfs had vanished; but the other man—he was now no higher than my knees—was standing by a tree behind the girl. He shouted again; and now the terror had left his face and he was grinning, I saw his hand go swiftly to his mouth. Had he taken more of his strange drug? Had he warned his two companions to do the same? I think so, for before my eyes he was swiftly diminishing in size. I knelt carefully beside the girl. Her figure—smaller than my foot and near it—was huddled into a little ball, her head against her upraised knees. She may have fainted; I did not heed her, save to be careful my movements did not strike her. With arm stretched over her I reached for the man. But he hopped away and eluded me. Still grinning. As small now as my little finger he stood half hiding behind a grass-blade. On hands and knees I pursued him. But like an insect, he was too quick for me. Smaller always until I was probing the grass with my fingers to find him—saw him momentarily like an ant in size as he leaped into a tangle of tiny grass-blades and was gone.

"I had forgotten my weapon. Illogically I had had no desire to kill that tiny figure—only to catch it. But Martt had had no such feelings. He was stamping around the glade—trying to stamp upon the other figures—and mumbling angrily to himself. I called to ask if he had caught them. He didn't know. He had seen them momentarily—seen them raise their hands to their mouths. But they had dwindled so fast, they were lost in a moment.

"The girl was unconscious, lying there in a huddled little heap. Gently I raised her, held her in the palm of my hand. She was white as a little waxen figure—white and beautiful; and so small I scarce dared to touch her with my huge rough fingers.

"Martt brought water from the lake. I rested my hand on the ground, with her still lying in it. And then presently she opened her eyes."

Brett paused, and as he gazed at each of us in turn I thought I had never seen his face so earnest. And there was upon it, too, a look almost of exaltation—a look which transfigured it. He added gently: "You three—my father, my sister, my friend, I have no need to hide from you my emotions. I think then—incongruously perhaps, for that little figure of girlhood lying there so soft and warm in the palm of my hand—I think then my love for her was born."

Hide his emotions! He could not had he wished. This love in his heart was written plain on his face, to soften it, to uplift it to something—or so it seemed to me—something just a little more than human. A touch, perchance, of divinity. And I think now that love does that—if only for some fleeting moment—to each one of us.

He went on very softly: "She opened her eyes. I was afraid she would be frightened. I tried to look very gentle, compassionate. I held my hand very still. I think that for an instant Martt and I stopped breathing. . . She opened her eyes—met mine. I saw in hers a flash of terror. But something, strangely, must have conquered it—against all reason as she stared at me. Stared while the terror faded, and her little lips parted and smiled a welcome and a thanks. . ."

CHAPTER 9

"DWINDLING GIANTS FROM LARGENESS UNFATHOMABLE"

There was not one of us who would have interrupted Brett when he paused to light an arrant-cylinder and to choose what next he would tell us. He was speaking softly, reminiscently, and with a curious gentleness.

"I carried her to the vehicle, showed it to her. Obviously she could understand nothing of my words; but she was very quick to read my gestures; smiling readily now, with her fear quite gone. And sitting up in the palm of my hand, with her arm flung about my thumb to steady her, she bade me raise her to my ear. Her words—the softest, the tiniest of human voices—what she said was wholly unintelligible, save that I understood her name was Leela.

"She stood beside a tree at a distance while we re-entered the vehicle and brought it down to a size normal to her; and came out of it to confront her."

Martt burst out: "I tell you that was when I realized how beautiful she is. Say, you never saw a girl like her—you can't describe it——"

"I'm not trying," said Brett with his gentle smile. "She met us—there by the vehicle—to us then, Frannie, she was about your size—perhaps a little smaller. She took our hands, laid them against her forehead as though with a gesture of welcome. And led us presently to her home—the house near by. Her father (her mother is dead) her father is a musician. Noted—very high of rank and standing among his people. A kindly old man, with gray and black hair worn long to the base of his neck. We—Martt and I—didn't let ours grow, though as you see we took their mode of dress."

"How long were you there?" I asked.

"We slept perhaps three hundred times," he answered. "There are no days and nights—always that same half-luminous twilight. No change of seasons—or very little. It is nature in her softest mood. Nothing to struggle against

—life made easy. Too easy. . . It was not we who learned Leela's language, but she, like an unnatural precocious child, who learned ours. . . We created a commotion among the people; the ruler sent for us. . . Oh, I have so much I'd like to tell you. But Martt can tell it—after——"

He checked himself suddenly. His words, some vague hint of what he almost had added, sent an ominous chill to my heart; and I saw, too, that Dr. Gryce had felt it, for a cloud came to his face and in his eyes I saw fear lurking.

But Brett went on at once: "I'd like to tell you of these people. A race at peace with nature and themselves. The struggle for existence all in the past. Decadence. The down-hill grade. Only by struggle can Man progress, Father. This race, with the peak of its civilization thousands of generations in its Past, gently resting, with the inevitable decadence drawing it inexorably back to the barbarism from whence it sprung. I'd like to tell you of their customs, their government—their mode of life. . . Some other time—or Martt will tell you. . . It was all so beautiful—so romantic. . . Music—their strange, beautiful arts—Music as Leela's father gave it—Art to take the place of Science and Industry. . . You ask Martt to tell you about the dancing—the pageants, if you want to call them that, to which we went so many times with Leela. . . But just now I'm tired—I think I've talked too much—and I'm worried—and it seems to press me, against all the logic of our Science, that I have no time to spend, telling all this to you. . ."

Brett, indeed, seemed suddenly tired, or perhaps harassed at the thoughts which had come to him. I had been so absorbed—as had all of us—that we had given no heed to the passing hours. Abruptly I realized that the room was chill with early morning; through the window I saw the flush of the eastern sky.

Martt followed my glance. "Why, it's dawn! Brett's been talking all night."

Brett said strangely: "Too long! Father, this gentle race living out there in such seeming security had just been visited by beings from the great world outside it. A world known to them only by legend of their past ages which they scarce knew to be true or false. Those three assailants of Leela's—and other men like them—had suddenly appeared as dwindling giants coming down out of largeness unfathomable. They had already destroyed a city. . ."

Brett's voice had risen; he was talking faster now; and there was a touch of wildness in his tone—a wildness perhaps born of his exhaustion, and the emotional stress under which I knew now he had been laboring all night.

"Our arrival there, Father—the three assailants of Leela—I think the larger, him whom we have called the 'giant'—I think he is leader of the invaders from that greater world. Our appearance—our own power to change size which perhaps he observed there in the forest—must have frightened him. The invaders vanished. But at the end of those months we lived there—another of these giants was seen.

"They're coming back again—to threaten Leela and all her people! I came here to see you, Father—to tell you all I've told—and to leave Martt. But I'm going back—to do what I can against this threat—this invasion. And I want to go back to Leela. She——"

"She was afraid to come with us," Martt put in. "I wanted her to come—and now I want to go back with Brett. We've been arguing about it for days—he won't let me go back with him—he's stubborn——"

Brett reiterated: "I'm going back. I'm going alone. As soon as I've slept—I've got to sleep now—you, you'll excuse me—let me take a good long sleep—I'm too tired to argue about it now. . . Good night, Frannie, dear—good night, Father—good night, Frank."

He was presently gone from the room. Dr. Gryce had been sitting beside me and I put my hand on his arm. His face was quite colorless; his voice, suddenly very old and helpless, was murmuring, "I don't want him to go out there again. I'm afraid—and I don't want him to do it. . ."

CHAPTER 10

THE SOLITARY VOYAGER

"But Brett," I said, "there are one or two things I want to ask you. About your return voyage—for instance——"

It was mid-afternoon. Brett, thoroughly rested, was wholly himself again. Quiet, composed and smiling, but very determined; even a little grim. And I think he was a bit ashamed of the sudden, almost querulous way in which he had terminated his narrative and left us there in the observation room at dawn. He had had his sleep now; and had been alone for an hour with his father. Martt and Frannie had been called to them; I—an outsider—was not asked, or wanted. What took place there behind the closed door of the study, it was not for me to ask. But when they came out I knew that Brett had won. A questionable victory, for old Dr. Gryce was visibly broken; Frannie—pale and upon the verge of tears; and Martt for a time a trifle sullen; resentful that he was to be left behind. I think it hurt Brett—this fear he was bringing upon those he loved. But he was very determined; convinced that it was the right thing for him to do.

"I start back tonight, Frank," he told me soberly as he emerged from the study.

"Oh," I said. "For how long will you be gone this time?"

He hesitated. A look, which even now my memory fails to interpret, came to him. Then he smiled. "I don't know. But remember, Frank, I can return—with only those limitations the Almighty enforces—I can return to any point of earth-Time I wish. As you will live it—well, I shall aim to return here within a month."

It was then I asked him about the return voyage he and Martt had just made. "Brett, I've been wondering—did our aural ray guide you back?"

"Yes," he said. "On the voyage back, the first thing I did was to put the vehicle back through Time to a chosen instant at which I wished to arrive here on earth. When that was done, I held that instant always. We could not see the aural ray going out—when we looked back for it—for two reasons. One: Our Time had run far into earth's Future, and the ray was non-existent. The other: Even had we taken the proper Time-point, we were outrunning the light-rays themselves. In space, I mean, the aural ray left earth only with the speed of light. Our velocity exceeded that. You see? But on the return voyage we encountered the ray as we came in. A mere flash over the sky; but its characteristic color-bands guided us."

What he said about outrunning the light-rays made me think of the myrdoscope, the image of that girl—which they had received here on earth before the voyage—that image had crossed a space 5,000,000 light-years in extent. But when I mentioned it, he explained:

"The myrdal rays are not light, Frank, but only akin to it. Their velocity—why, light beside them is a laggard. We have no way of computing the velocity of the myrdal rays. But over a finite distance such as five million light-years—for practical purposes it is instantaneous. . .

"I wanted to tell you—I was confused last night—I meant to explain that coming back I used quite a different method from the outward trip. I chanced a disturbance of some of those outlying starry universes, and when we left the Inner Surface, I made the vehicle larger instead of smaller. The void of Space shrank until about us the universes were clustered like little patches of mist—tiny areas of glowing star-dust. I saw our own, with its spectrum of the aural ray, quite readily. And had reached it with a voyage of a few hours—and then reduced our size."

"And your Time," I said. "Brett, I didn't see the vehicle until it was almost entering the earth's atmosphere. And—just for an instant—it seemed not solid, but like a vague gray ghost. Then suddenly it materialized."

He smiled and nodded. "Yes. That was when I took the earth's normal Time-rate."

The family joined us; we said no more. And that night Brett left us for his solitary voyage. I would not set down here in detail those last good-byes. Emotion repressed—it was what was not said that held a pathos I shall never forget. An outward attempt at lightness. Martt laughed, "Give my love to Leela." And Frannie said, "You tell her I'm jealous because she's so beautiful."

Just before Brett closed the door of the vehicle, Dr. Gryce spoke—the only thing he had said for an hour past.

"You'll be sure to come back, Brett? Within the month, lad?"

"Oh, yes. Yes, Father dear."

"Well—good-bye. . ."

Good-bye! I can think of no sadder word for human tongue to frame.

CHAPTER 11

BRAVE LITTLE BEACON STRIVING TO PIERCE INFINITY

That little month of anxious watching and waiting passed so slowly! And yet so quickly, as one by one its golden moments of hope drained away.

Brett did not return. A month, then a year, while Dr. Gryce made me leave the Service, to enter his, that all my time might be spent in watching.

A year; and now another year has passed. Brett would return within the month. With his Time-mechanism unimpaired, no delay out there in the Beyond could have affected his return to reach us during that first little month. With that passed and gone, reason could only show the futility of expecting him ever. Yet reason plays so small a part, when it would seek to kill hope.

The aural ray still burns—brave little beacon striving to pierce infinity. Beside it, for those long, unreasoning hours of vigil, Dr. Gryce sits and waits; silent, grayer and every day visibly older. The possibilities of what could have happened to Brett—that myriad of futile human conjectures—we have long since ceased voicing. Alone, I sometimes speculate. Has Brett gone on into that outside world of which we all are only a tiny atom? What is he doing? And then I tell myself, what is it to me, save that it concerns Brett? The myriad, unfathomable happenings of Eternal Time in Infinite Space—what right have I, one tiny mortal, to probe them?

The beacon burns to guide Brett back to us. Will he ever come? I wonder. My brain, with its logic, says he will not. But my heart says, "Might he not come tonight?" Or with tonight passed, then tomorrow he will be here. Thus hope runs on and on, daunted but never broken. Blessed hope, to make possible a courageous living of this little life until we ourselves are plunged into that glowing Infinity of the Hereafter.

THE END

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